

Chapter 1: Introduction

1.1 Introduction

In today's digital world, almost every company depends on a **computer network** for communication and daily work. A network helps employees share files, access data, send emails, use company software, and communicate with branch offices. A secure and reliable network is very important for smooth business operations.

For example, imagine a company like a bank, hospital, shopping company, or IT company that has a **Head Office (HQ)** and many branch offices in different cities. Employees from departments such as **Manager, IT, HR, and Sales** need to communicate and share information regularly. If the network is slow or fails, employees may not access important files, customer data, or office resources, which can affect business operations.

In many organizations, different departments perform different tasks. For example:

- The **HR Department** manages employee records.
- The **Sales Department** handles customer and business activities.
- The **IT Department** manages technical systems and network issues.
- The **Manager Department** monitors company operations.

If all departments use the same network without separation, it may create **high traffic, low performance, and security problems**. For example, if a problem occurs in one department, it may affect the whole network. Therefore, companies use **VLAN (Virtual Local Area Network)** to separate departments into different networks for better performance and security.

Another common problem in companies is communication between multiple offices. Suppose a company has a **Head Office in Delhi** and branch offices in **Noida and Gurgaon**. These offices need continuous communication for sharing reports, customer details, and office data. If routing is not properly configured, communication can fail. To solve this issue, companies use **OSPF (Open Shortest Path First)**, which automatically finds the best path for communication between offices.

Network failure is also a major problem in organizations. For example, if the main router in a company suddenly stops working, employees may lose network access, causing business interruption. To avoid this problem, companies use **HSRP (Hot Standby Router Protocol)**, which provides a backup router. If one router fails, another router automatically takes over and keeps the network running.

Large networks often contain multiple switches to provide better connectivity. However, connecting multiple switches may create **network loops**, which can slow down or crash the network. To solve this problem, organizations use **STP (Spanning Tree Protocol)**, which prevents switching loops and keeps the network stable.

Companies also need secure internet access for emails, online meetings, and cloud services. Internal company networks mostly use **private IP addresses**, which cannot directly access the internet. Therefore, organizations use **NAT (Network Address Translation)** to convert private IP addresses into public IP addresses for secure internet communication.

This project, “**Enterprise Network Simulation using Cisco Packet Tracer**”, is developed to simulate a **real-life enterprise network environment**. In this project, a **Headquarters (HQ)** is connected with **two Branch Offices** using networking technologies such as **VLAN, OSPF, HSRP, STP, and NAT**. The network includes departments like **HQ Manager, IT Department, HR Department, and Sales Department** to simulate a real company structure.

The project is designed using **Cisco Packet Tracer**, which is a network simulation software used to configure and test networking devices such as **routers, switches, servers, and PCs**. This helps students understand how enterprise networks work in real organizations without using expensive hardware devices.

This project is useful for learning **enterprise networking concepts, routing, network security, redundancy, and communication management**. It also provides practical knowledge of how companies maintain communication between departments and branch offices securely and efficiently.

Branch Office	VLAN ID	IP Address Range	Connected To	Purpose
Branch Office 1	50	192.168.50.0/24	Headquarters (HQ)	Communication and data sharing between HQ and Branch Office 1
Branch Office 2	60	192.168.60.0/24	Headquarters (HQ)	Communication and data sharing between HQ and Branch Office 2

Table 1: Branch Office Network Details

1.2 Importance of the Project

This project is important because it helps in understanding how a **real company network** works. It provides practical knowledge of **routing, switching, redundancy, and network security**.

In real organizations, a proper network helps:

- Improve communication between departments and branches
- Reduce network failures
- Improve security and data sharing
- Maintain smooth business operations
- Provide reliable internet connectivity

This project also helps students understand **real-world enterprise networking**, which is useful for careers in **Networking, System Administration, IT Support, and Cybersecurity**.

Chapter 2: Literature Review

2.1 : Introduction

A literature review is the study of existing technologies, concepts, and methods related to the project. It helps in understanding how similar systems work and what technologies are used in real-world environments.

This project, “**Enterprise Network Simulation using Cisco Packet Tracer**”, is based on enterprise networking concepts such as **VLAN, OSPF, HSRP, STP, and NAT**. These technologies are commonly used in companies, banks, hospitals, educational institutions, and IT organizations to create secure and reliable networks.

2.2 : Enterprise Networking

Enterprise networking is used to connect different departments and branch offices of an organization. A proper enterprise network allows employees to communicate, share files, access servers, and use internet services securely.

For example, a company with a **Head Office (HQ)** and branch offices in different cities requires a network for communication and data sharing. Without a proper network, employees may face communication problems and data transfer delays.

Enterprise networks are designed to provide:

- Secure communication
- Fast data transfer
- High network availability
- Better performance
- Scalability for future growth

2.3 : VLAN (Virtual Local Area Network)

VLAN (Virtual Local Area Network) is used to divide a network into smaller logical groups. In companies, different departments are separated using VLANs to improve security and reduce unnecessary traffic.

For example:

- **VLAN 10 → HQ Manager Department**
- **VLAN 20 → IT Department**
- **VLAN 30 → HR Department**
- **VLAN 40 → Sales Department**

Without VLANs, all devices remain in one network, which may reduce performance and create security issues.

VLAN ID	Department Name	IP Network	Purpose
10	HQ Manager Department	192.168.10.0/24	Used for management and monitoring activities
20	IT Department	192.168.20.0/24	Used for technical support and network management
30	HR Department	192.168.30.0/24	Used for employee records and HR activities
40	Sales Department	192.168.40.0/24	Used for customer and sales-related activities
50	Branch Office 1	192.168.50.0/24	Used for communication with Branch Office 1
60	Branch Office 2	192.168.60.0/24	Used for communication with Branch Office 2

Table 2: VLAN and Department Details

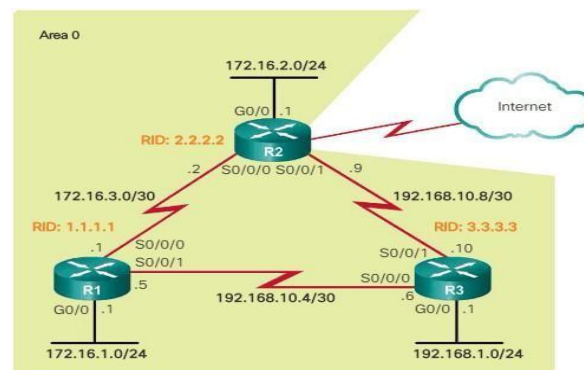
2.4 OSPF (Open Shortest Path First)

OSPF (Open Shortest Path First) is a dynamic routing protocol used to connect different offices and networks.

For example, if a company has offices in **Delhi, Noida, and Gurgaon**, OSPF automatically finds the best route for communication. If one route fails, OSPF selects another available path automatically.

Benefits of OSPF:

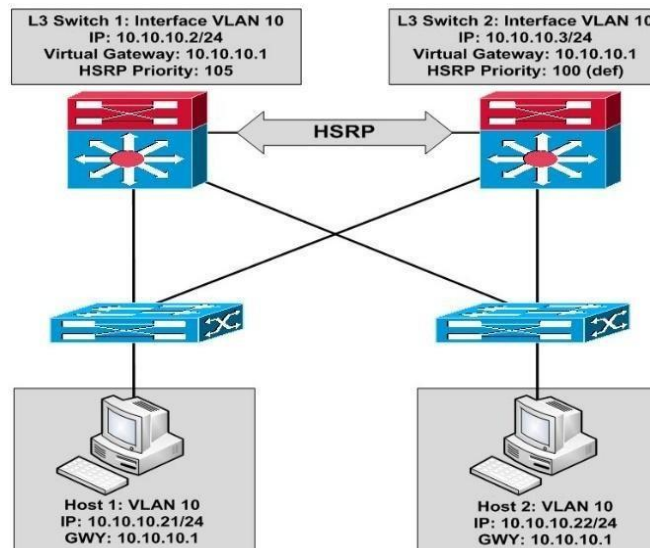
- Fast communication
- Automatic route updates
- Better network management
- Suitable for large networks



2.5 : HSRP (Hot Standby Router Protocol)

HSRP (Hot Standby Router Protocol) is used for network redundancy and backup.

In a company, if the main router stops working, employees may lose internet or network access. HSRP solves this problem by keeping a backup router ready.



For example:

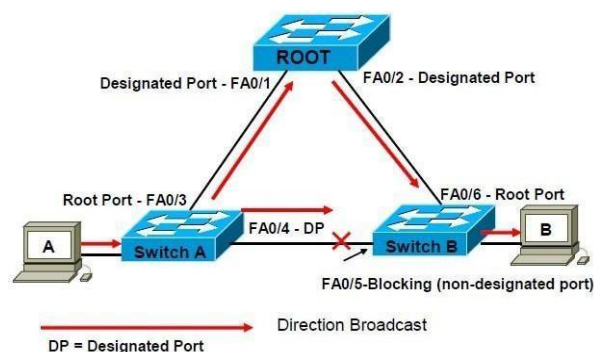
- **Router 1 = Active Router (Priority 100)**
- **Router 2 = Standby Router (Priority 105)**

If Router 1 fails, Router 2 automatically takes control without affecting communication.

2.6 : STP (Spanning Tree Protocol)

STP (Spanning Tree Protocol) prevents switching loops in the network. In enterprise networks, multiple switches are connected for better communication. However, redundant links can create loops and crash the network.

Final STP Topology



STP blocks unnecessary paths and keeps only the best route active.

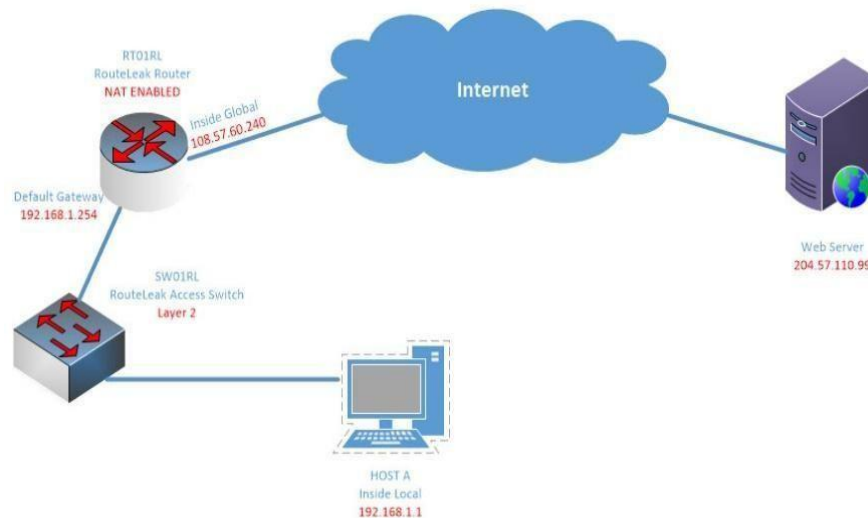
Benefits of STP:

- Prevents loops
- Improves network stability
- Reduces broadcast traffic

2.7 : NAT (Network Address Translation)

NAT (Network Address Translation) is used to provide secure internet access.

In companies, devices use private IP addresses, which cannot directly connect to the internet. NAT converts private IP addresses into public IP addresses.



For example:

- Private IP: **192.168.10.10**
- Public IP: **Internet Accessible Address**

Benefits of NAT:

- Secure internet communication
- Saves IP addresses
- Protects internal network

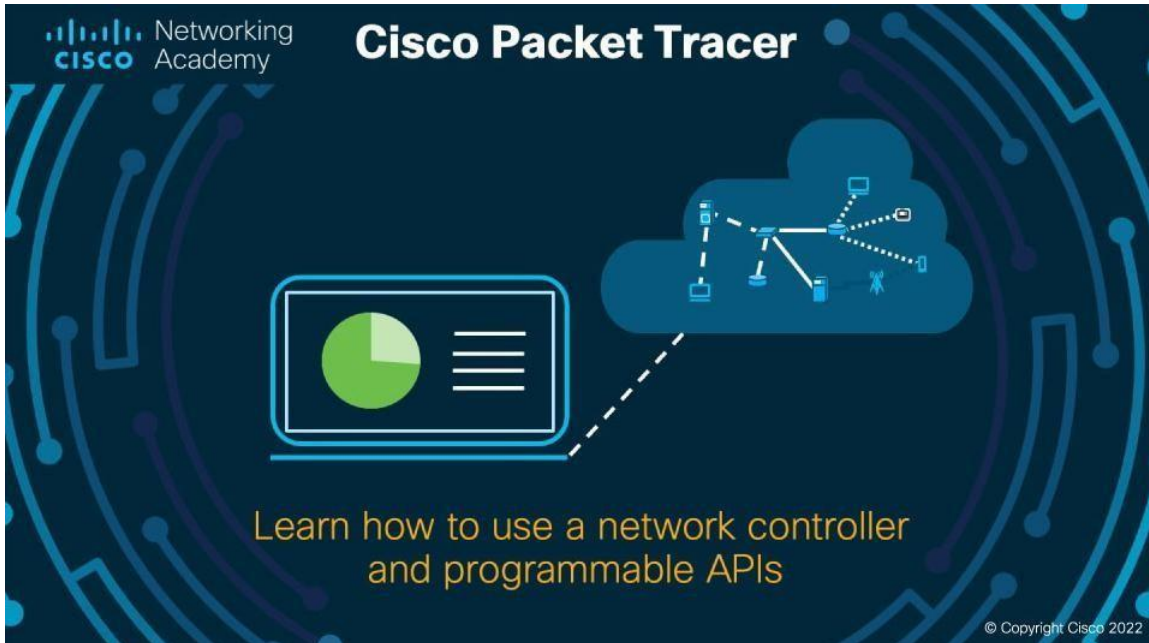
2.8 : Cisco Packet Tracer

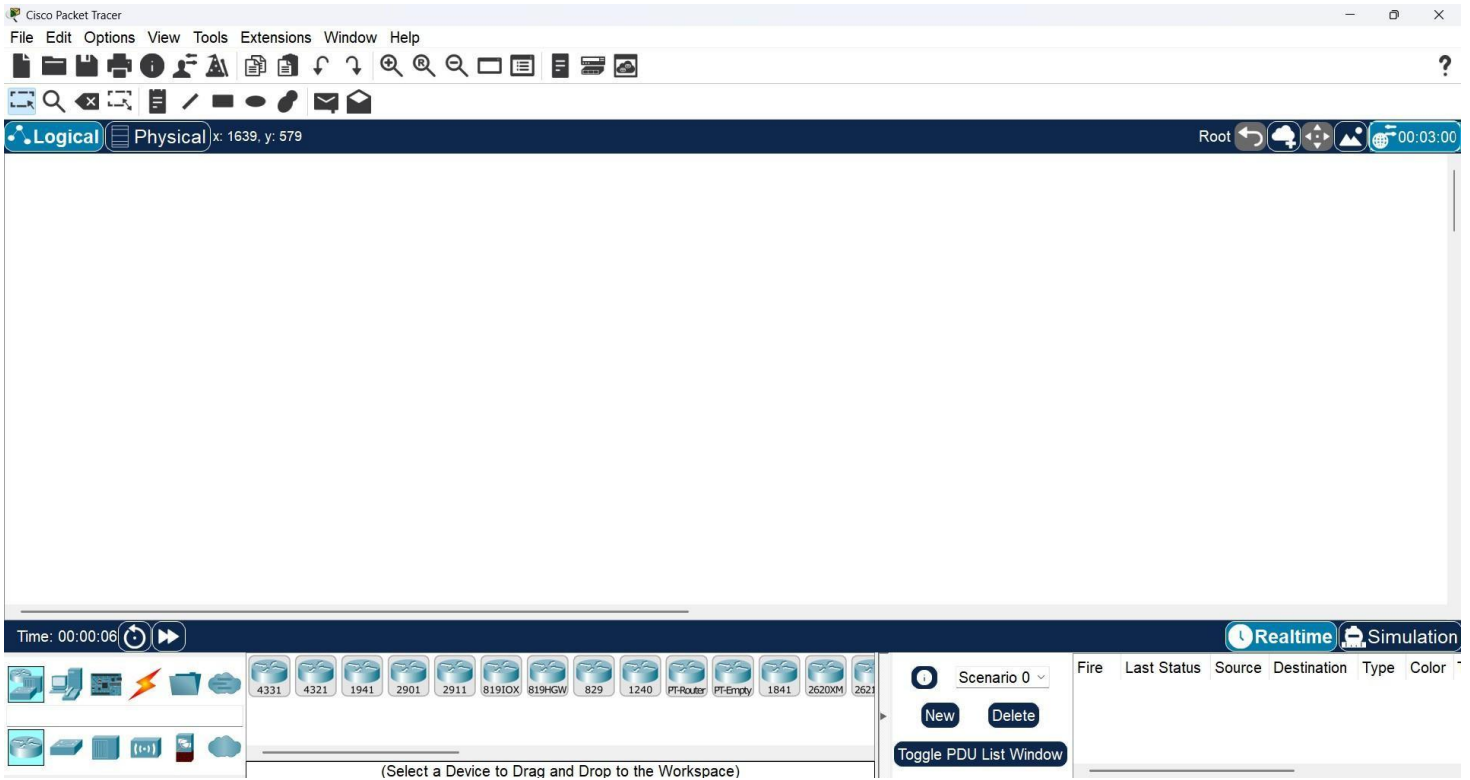
Cisco Packet Tracer is a network simulation software used to design and test networking projects.

It helps students and network engineers:

- Configure routers and switches
- Test network communication
- Learn routing and switching practically
- Simulate enterprise networks without real hardware

In this project, Cisco Packet Tracer is used to simulate the **Headquarters (HQ)** and **Branch Offices** network.





2.9 Summary

From the literature review, it is clear that technologies like **VLAN, OSPF, HSRP, STP, and NAT** play an important role in building a secure and reliable enterprise network. These technologies help organizations maintain communication, improve security, and avoid network failures. The study of these concepts helps in designing an effective enterprise network system for real-world use.

Chapter 3: System Analysis & Design

3.1 Introduction

System Analysis and Design is an important step in project development. It helps in understanding the requirements, planning the network structure, and designing a proper system for smooth communication. In this project, an **Enterprise Network** is designed to connect a **Headquarters (HQ)** with **two Branch Offices** using secure and reliable networking technologies.

The system is designed to provide **better communication, network security, redundancy, and internet connectivity** between departments and offices.

3.2 Existing System

In traditional networks, all devices are connected in a single network without proper management. This creates many problems such as:

- High network traffic
- Poor communication speed
- No department separation
- Security issues
- Single point of failure
- Difficult network management

For example, if all departments like **HR, IT, Sales, and Manager** are connected in one network, unnecessary traffic may reduce performance and affect communication.

3.3 Proposed System

The proposed system is an **Enterprise Network Simulation using Cisco Packet Tracer** where a **Headquarters (HQ)** is connected with **Branch Office 1** and **Branch Office 2**.

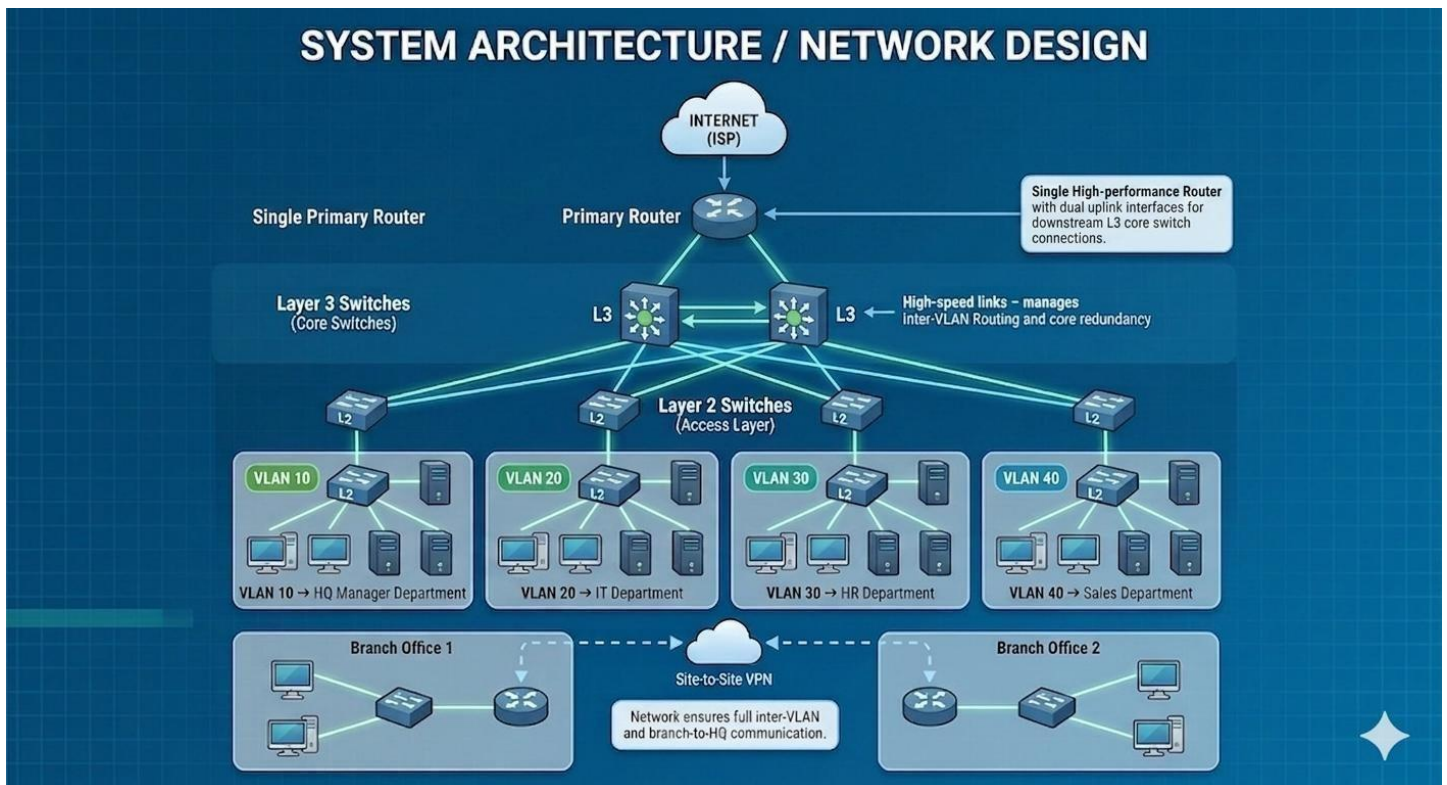
The proposed network includes:

- **VLAN (Virtual Local Area Network)** for department separation
- **OSPF (Open Shortest Path First)** for routing between offices
- **HSRP (Hot Standby Router Protocol)** for backup gateway
- **STP (Spanning Tree Protocol)** to prevent loops
- **NAT (Network Address Translation)** for secure internet access

This system provides better performance, security, and network availability.

3.4 System Architecture / Network Design

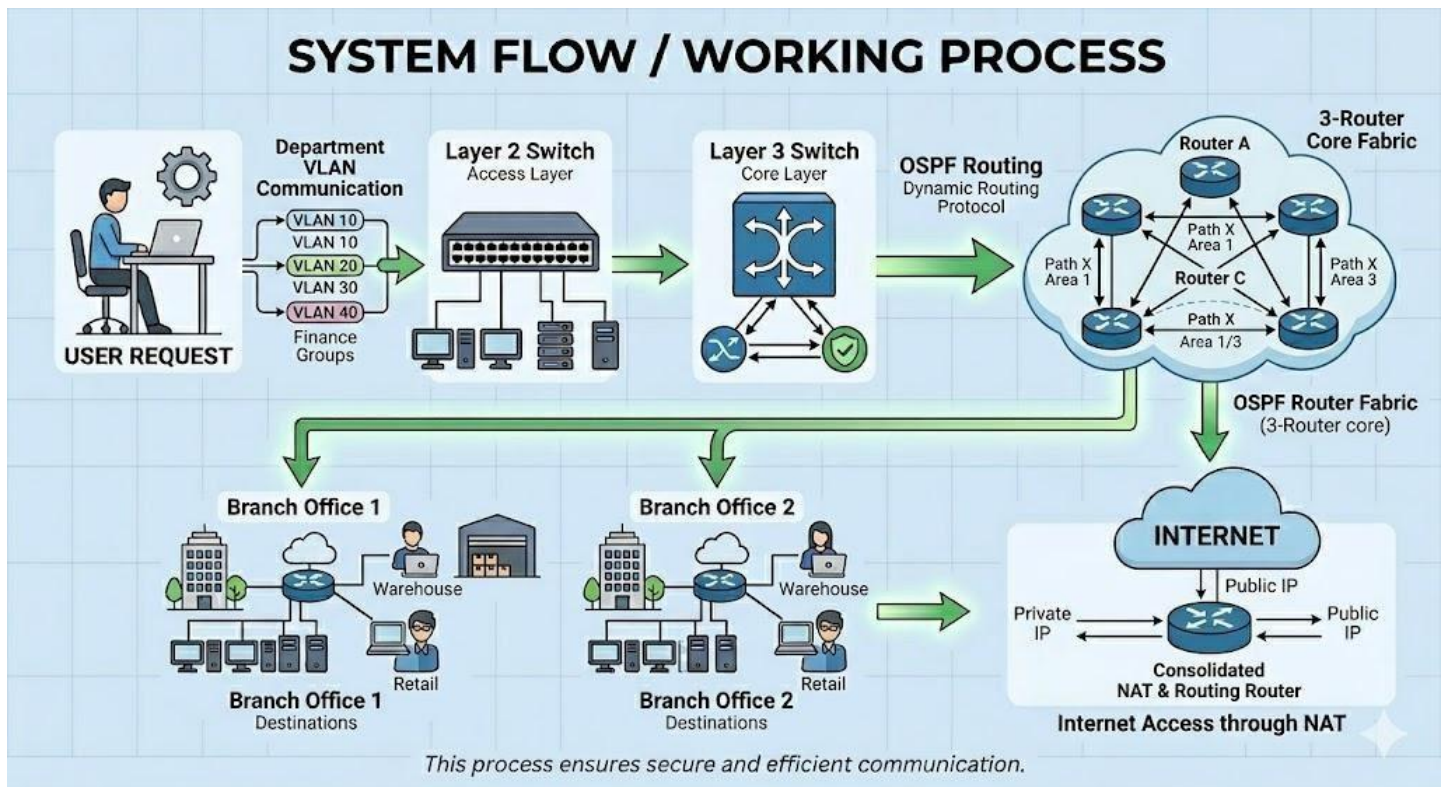
The network architecture is designed in a hierarchical structure:



The network ensures communication between all departments and branch offices.

3.5 System Flow / Working Process

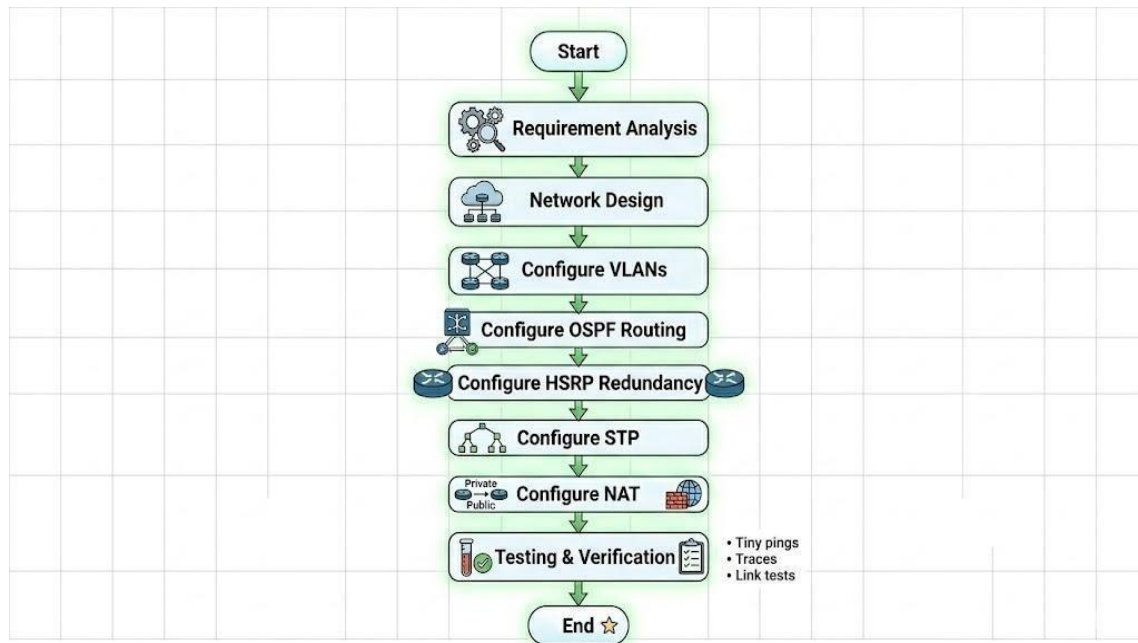
This process ensures secure and efficient communication.



3.6 IP Addressing Scheme

Department / Office	VLAN ID	IP Address Range
IT Department	20	192.168.20.0/24
HR Department	30	192.168.30.0/24
Sales Department	40	192.168.40.0/24
Branch Office 1	50	192.168.50.0/24
Branch Office 2	60	192.168.60.0/24

3.7 Flowchart Start



3.8 Advantages of the Proposed System

- Better communication between departments
- Secure data sharing
- Reduced network traffic using VLAN
- Backup connectivity using HSRP
- Loop prevention using STP
- Secure internet access using NAT
- Scalable network for future expansion

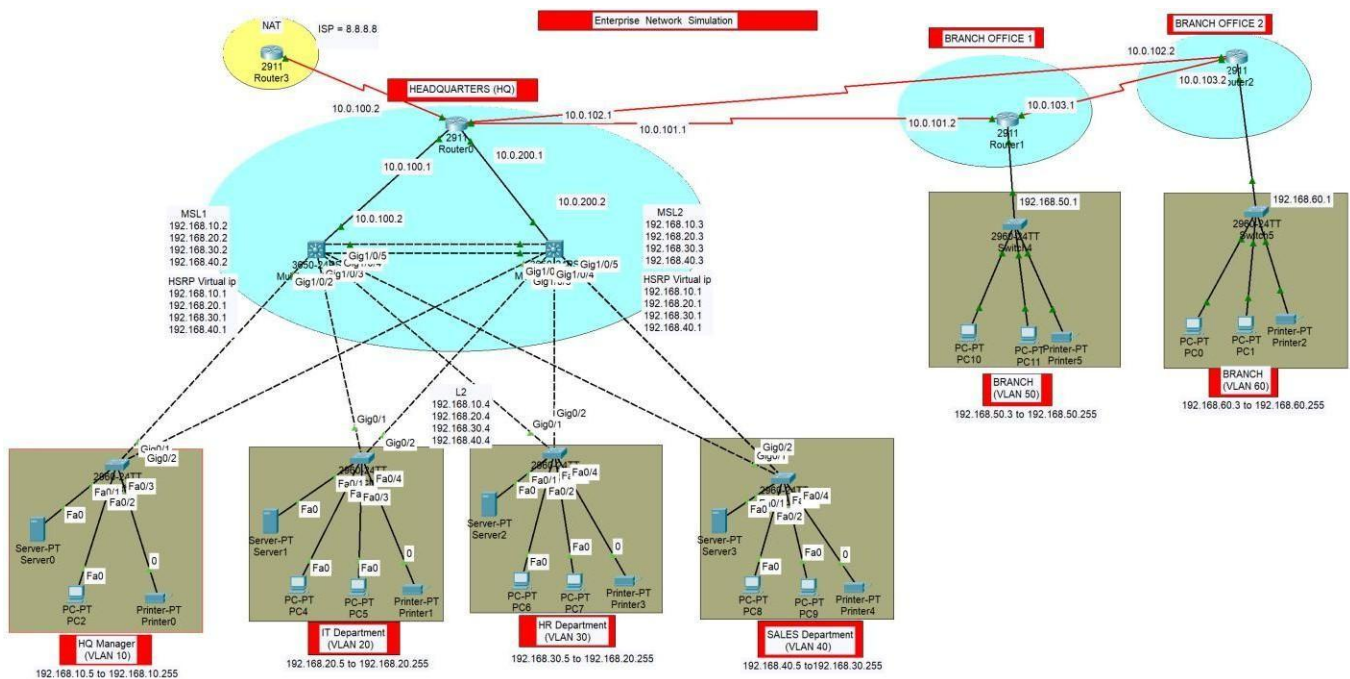
3.9 Summary

The proposed enterprise network is designed to provide **secure, reliable, and scalable communication** between the Headquarters and Branch Offices. By using technologies like **VLAN, OSPF, HSRP, STP, and NAT**, the network improves performance, security, and availability in a real-world enterprise environment.

Chapter 4: Implementation

4.1 Introduction

Implementation is the process of designing, configuring, and testing the enterprise network in **Cisco Packet Tracer**. In this phase, all required devices such as **routers, switches, PCs, and servers** are configured to establish communication between the **Headquarters (HQ)** and **two Branch Offices**.



The implementation focuses on creating a **secure, reliable, and scalable enterprise network** using technologies such as **VLAN, OSPF, HSRP, STP, and NAT**.

S. No.	Hardware Component	Specification / Model	Purpose
1	Router	Cisco 2911 Router	Used for routing between HQ and Branch Offices
2	Multilayer Switch	Layer 3 Switch	Used for VLAN routing and HSRP configuration
3	Switch	Layer 2 Switch	Used for connecting PCs and devices
4	Personal Computer (PC)	Standard Desktop PC	Used for communication and testing
5	Server	Generic Server	Used for data sharing and network services
6	Printer	Network Printer	Used for printing documents in the network
7	Ethernet Cable	Copper Straight-through Cable	Used for device connectivity

Table 5&6: Hardware Requirements & Devices Used in the Network

S. No.	Software Name	Version / Type	Purpose
1	Cisco Packet Tracer	Latest Version	Used to design, configure, and simulate the enterprise network
2	Windows Operating System	Windows 10/11	Used for running Cisco Packet Tracer and project development
3	Microsoft Word	MS Office / Word	Used for documentation and report preparation
4	Web Browser	Google Chrome / Edge	Used for research and accessing networking resources

Table 6: Software Requirements

4.2 Network Setup

The enterprise network is designed in **Cisco Packet Tracer** using the following devices:

Devices Used

- **Cisco 2911 Routers**



- **Layer 3 (Multilayer) Switches**



- **Layer 2 Switches**



- **PCs (Personal Computers)**



- Servers

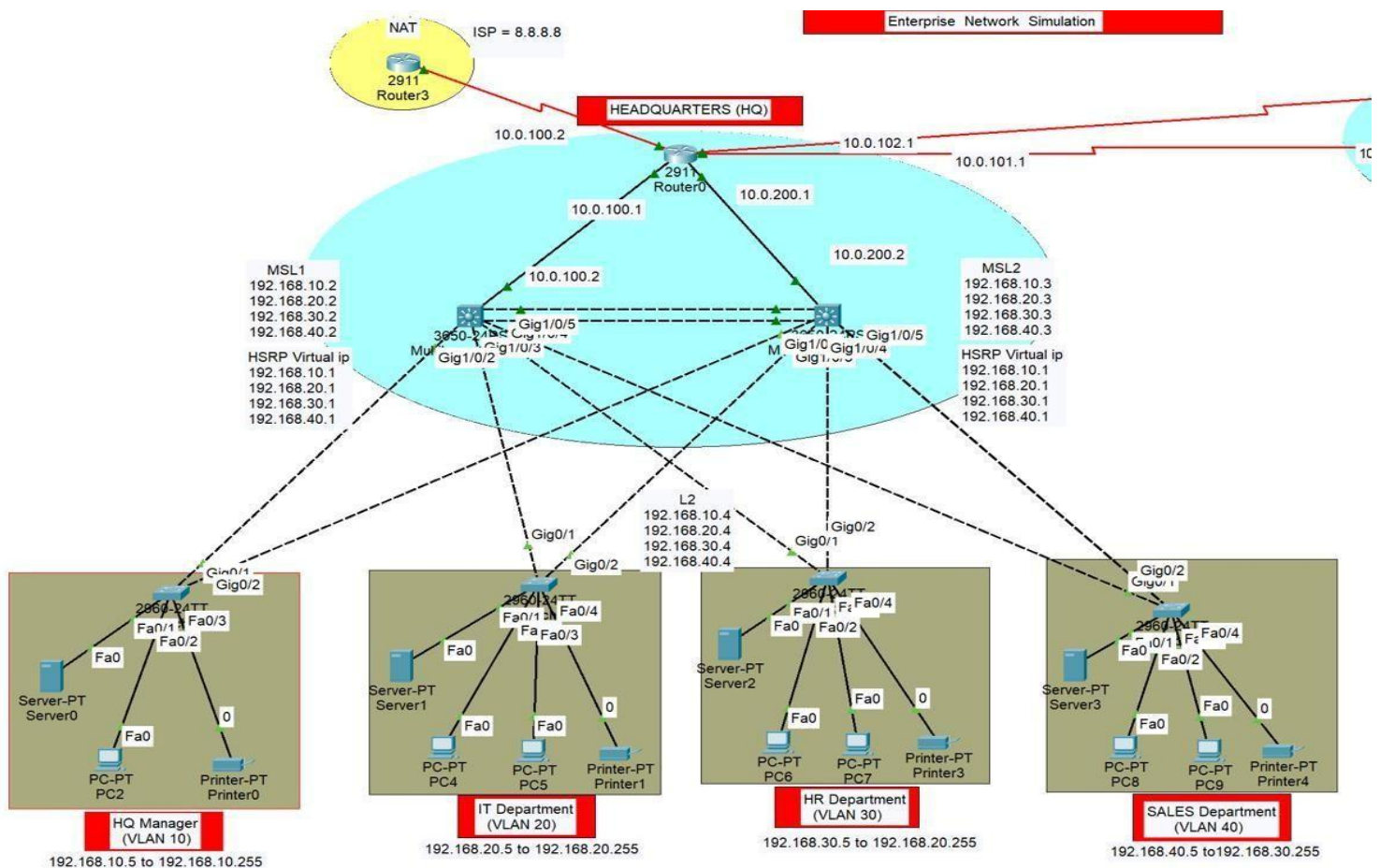


- Printers

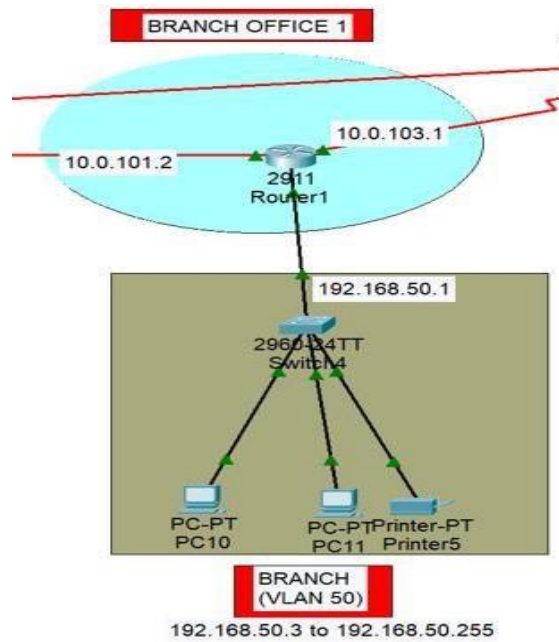


The network includes:

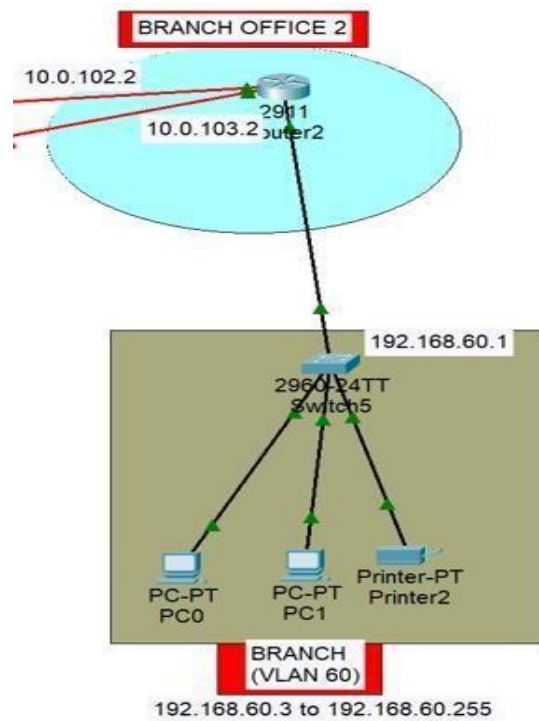
- Headquarters (HQ)



- Branch Office 1



- Branch Office 2



4.3 VLAN Configuration

VLAN (Virtual Local Area Network) is implemented to separate departments into different networks for better security and traffic management.

- Multiple departments using VLANs

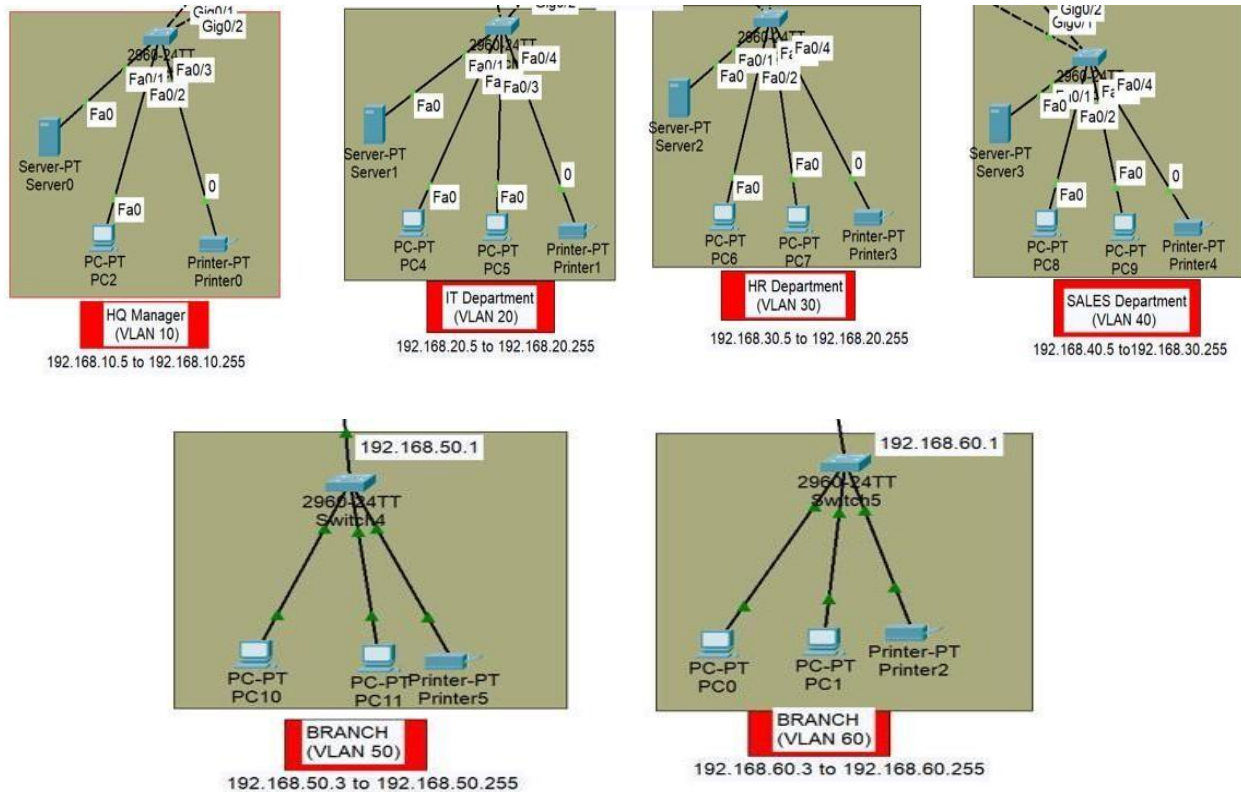


FIGURE 4.3: VLAN Configuration

4.3.1 : VLAN 10(HQ Manager) switch configuration

```
HQswitch>enable
```

```
HQswitch#config terminal
```

```
HQswitch(config)# vlan 10
```

```
HQswitch(config-vlan)# name HR_Manager
```

```
HQswitch(config-vlan)# exit
```

```
HQswitch(config)# interface vlan 10
```

```
HQswitch(config-if)# ip address 192.168.10.5 255.255.255.0
```

```
HQswitch(config-if)# no shutdown
HQswitch(config-if)# exit
```

```
HQswitch(config)# interface range fa0/1-24
HQswitch(config-if-range)# switchport mode access
HQswitch(config-if-range)# switchport access vlan 10
HQswitch(config-if-range)# exit
HQswitch(config)#do write
Building configuration...
[OK]
```

```
HQSwitch>enable
HQSwitch#config terminal
```

```
HQSwitch(config)#interface range gig0/1-2
HQSwitch(config-if)#switchport mode trunk
HQSwitch(config-if)#no shutdown
HQSwitch(config-if)#exit
```

```
HQSwitch(config)#do write
Building configuration...
[OK]
```

4.3.2 : VLAN 20(IT_Department) switch configuration

```
IT_Department>enable
IT_Department#config terminal
```

```
IT_Department(config)# vlan 20
IT_Department(config-vlan)# name IT_Department
IT_Department(config-vlan)# exit
```

```
IT_Department(config)# interface vlan 20
IT_Department(config-if)# ip address 192.168.20.5 255.255.255.0
IT_Department(config-if)# no shutdown
IT_Department(config-if)# exit
```

```
IT_Department(config)# interface range fa0/1-24
IT_Department(config-if-range)# switchport mode access
IT_Department(config-if-range)# switchport access vlan 20
```

```
IT_Department(config-if-range)# exit
IT_Department(config)#do write
Building configuration...
[OK]
```

```
IT_Department>enable
```

```
IT_Department#config terminal
```

```
IT_Department(config)#interface range gig0/1-2  
IT_Department(config-if)#switchport mode trunk  
IT_Department(config-if)#no shutdown  
IT_Department(config-if)#exit
```

```
IT_Department(config)#do write  
Building configuration...  
[OK]
```

4.3.3 : VLAN 30(HR_Department) switch configuration

```
HR_Department>enable  
HR_Department#config terminal
```

```
HR_Department(config)# vlan 30  
HR_Department(config-vlan)# name HR_Department  
HR_Department(config-vlan)# exit
```

```
HR_Department(config)# interface vlan 30  
HR_Department(config-if)# ip address 192.168.30.5 255.255.255.0  
HR_Department(config-if)# no shutdown  
HR_Department(config-if)# exit
```

```
HR_Department(config)# interface range fa0/1-24  
HR_Department(config-if-range)# switchport mode access  
HR_Department(config-if-range)# switchport access vlan 30  
HR_Department(config-if-range)# exit
```

```
HR_Department(config)# do write  
Building configuration...  
[OK]
```

```
HR_Department>enable  
HR_Department#config terminal
```

```
HR_Department(config)#interface range gig0/1-2  
HR_Department(config-if)#switchport mode trunk  
HR_Department(config-if)#no shutdown  
HR_Department(config-if)#exit
```

```
HR_Department(config)#do write  
Building configuration...  
[OK]
```

4.3.4 : VLAN 40(Sales_Department) switch configuration

```
Sales_Department>enable
Sales_Department#config terminal

Sales_Department(config)# vlan 40
Sales_Department(config-vlan)# name Sales_Department
Sales_Department(config-vlan)# exit

Sales_Department(config)# interface vlan 40
Sales_Department(config-if)# ip address 192.168.40.5 255.255.255.0
Sales_Department(config-if)# no shutdown
Sales_Department(config-if)# exit

Sales_Department(config)# interface range fa0/1-24
Sales_Department(config-if-range)# switchport mode access
Sales_Department(config-if-range)# switchport access vlan 40
Sales_Department(config-if-range)# exit

Sales_Department(config)# do write
Building configuration...
[OK]
Sales_Department>enable
Sales_Department#config terminal

Sales_Department(config)#interface range gig0/1-2
Sales_Department(config-if)#switchport mode trunk
Sales_Department(config-if)#no shutdown
Sales_Department(config-if)#exit

Sales_Department(config)#do write
Building configuration...
[OK]
```

4.3.5 : VLAN 50 (Branch Office 1) switch configuration

```
Branch_1>enable
Branch_1#config terminal

Branch_1(config)# vlan 50
Branch_1(config-vlan)# name Branch_1
Branch_1(config-vlan)# exit

Branch_1(config)# interface vlan 50
Branch_1(config-if)# ip address 192.168.50.3 255.255.255.0
Branch_1(config-if)# no shutdown
```

```
Branch_1(config-if)# exit
Branch_1(config)# interface range fa0/1-24
Branch_1(config-if-range)# switchport mode access
Branch_1(config-if-range)# switchport access vlan 50
Branch_1(config-if-range)# exit
```

```
Branch_1(config)# do write
Building configuration...
[OK]
```

```
Branch_1>enable
Branch_1#config terminal
```

```
Branch_1(config)#interface range gig0/1-2
Branch_1(config-if)#switchport mode trunk
Branch_1(config-if)#no shutdown
Branch_1(config-if)#exit
```

```
Branch_1(config)#do write
Building configuration...
[OK]
```

4.3.6 : VLAN 60 (Branch Office 2) switch configuration

```
Branch_2>enable
Branch_2#config terminal
```

```
Branch_2(config)# vlan 60
Branch_2(config-vlan)# name Branch_2
Branch_2(config-vlan)# exit
```

```
Branch_2(config)# interface vlan 60
Branch_2(config-if)# ip address 192.168.60.4 255.255.255.0
Branch_2(config-if)# no shutdown
Branch_2(config-if)# exit
```

```
Branch_2(config)# interface range fa0/1-24
Branch_2(config-if-range)# switchport mode access
Branch_2(config-if-range)# switchport access vlan 60
Branch_2(config-if-range)# exit
```

```
Branch_2(config)# do write
Building configuration...
[OK]
```

```
Branch_2>enable
```

```
Branch_2#config terminal
```

```
Branch_2(config)#interface range gig0/1-2  
Branch_2(config-if)#switchport mode trunk  
Branch_2(config-if)#no shutdown  
Branch_2(config-if)#exit
```

```
Branch_2(config)#do write  
Building configuration...  
[OK]
```

4.3.7 : Configured VLANs

Department	VLAN ID	IP Network
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Department	VLAN ID	IP Network
HQ Manager Department	10	192.168.10.5 to 192.168.10.255/24
IT Department	20	192.168.20.5 to 192.168.20.255/24
HR Department	30	192.168.30.5 to 192.168.30.255/24
Sales Department	40	192.168.40.5 to 192.168.40.255/24
Branch Office 1	50	192.168.50.3 to 192.168.50.255/24
Branch Office 2	60	192.168.60.3 to 192.168.60.255/24

This configuration helps reduce unnecessary traffic and improves security.

4.4 OSPF Configuration

OSPF (Open Shortest Path First) is configured to establish communication between the **Headquarters (HQ)** and **Branch Offices**.

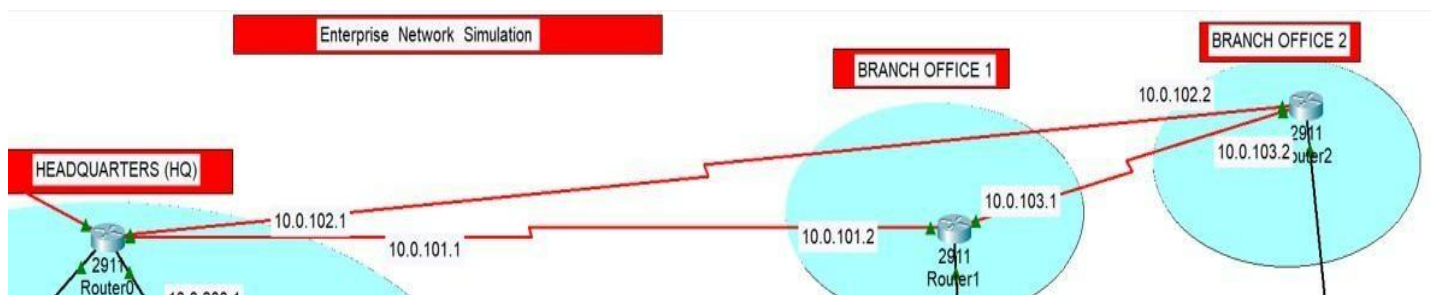


FIGURE 4.4: OSPF Configuration

Purpose of OSPF

- Dynamic routing between offices
- Automatic route updates
- Fast communication between locations

After configuration, all branch offices successfully communicate with HQ.

4.4.1 : HQ Router (HQ ↔ BR1, HQ ↔ BR2)

```
HQ_Router>enable
HQ_Router#config terminal

HQ_Router(config)#interface s0/0/0
HQ_Router(config-if)#ip address 10.0.101.1 255.255.255.248
HQ_Router(config-if)#no shutdown
HQ_Router(config-if)#exit

HQ_Router(config)#interface s0/0/1
HQ_Router(config-if)#ip address 10.0.102.1 255.255.255.248
HQ_Router(config-if)#no shutdown
HQ_Router(config-if)#exit
HQ_Router(config)#end
```

4.4.1.1 : HQ OSPF Configuration

```
HQ_Router>enable
HQ_Router#config terminal

HQ_Router(config)#router ospf 1
HQ_Router(config-router)#router-id 1.1.1.1

HQ_Router(config-router)#network 10.0.101.0 0.0.0.7 area 0
HQ_Router(config-router)#network 10.0.102.0 0.0.0.7 area 0

HQ_Router(config-router)#exit
HQ_Router(config)#end
HQ_Router#
```

4.4.2 : BR1 Router (BR1 ↔ HQ, BR1 ↔ BR2)

```
BR1_Router>enable
BR1_Router#config terminal

BR1_Router(config)#interface s0/0/0
BR1_Router(config-if)#ip address 10.0.101.2 255.255.255.248
BR1_Router(config-if)#no shutdown
BR1_Router(config-if)#exit

BR1_Router(config)#interface s0/0/1
BR1_Router(config-if)#ip address 10.0.103.1 255.255.255.248
BR1_Router(config-if)#no shutdown
```

```
BR1_Router(config-if)#exit
```

```
BR1_Router(config)#end
```

```
BR1_Router#
```

```
BR1_Router>enable
```

```
BR1_Router#config terminal
```

4.4.2.1 : BR1_Router OSPF Configuration

```
BR1_Router(config)#router ospf 1
```

```
BR1_Router(config-router)#router-id 2.2.2.2
```

```
BR1_Router(config-router)#network 10.0.101.0 0.0.0.7 area 0
```

```
BR1_Router(config-router)#network 10.0.103.0 0.0.0.7 area 0
```

```
BR1_Router(config-router)#exit
```

```
BR1_Router(config)#end
```

```
BR1_Router#
```

4.4.3 : BR2 Router (BR2 ↔ HQ, BR2 ↔ BR1)

```
BR2_Router>enable
```

```
BR2_Router#config terminal
```

```
BR2_Router(config)#interface s0/0/0
```

```
BR2_Router(config-if)#ip address 10.0.102.2 255.255.255.248
```

```
BR2_Router(config-if)#no shutdown
```

```
BR2_Router(config-if)#exit
```

```
BR2_Router(config)#interface s0/0/1
```

```
BR2_Router(config-if)#ip address 10.0.103.2 255.255.255.248
```

```
BR2_Router(config-if)#no shutdown
```

```
BR2_Router(config-if)#exit
```

```
BR2_Router(config)#end
```

4.4.3.1 : BR2_Router OSPF Configuration

```
BR2_Router>enable
```

```
BR2_Router#config terminal
```

```
BR2_Router(config)#router ospf 1
```

```
BR2_Router(config-router)#router-id 3.3.3.3
```

```
BR2_Router(config-router)#network 10.0.102.0 0.0.0.7 area 0
```

```
BR2_Router(config-router)#network 10.0.103.0 0.0.0.7 area 0
```

```
BR2_Router(config-router)#exit
```

4.4.4 : HQ_Router OSPF (MLS1 + MLS2)

```
HQ_Router>enable
```

```
HQ_Router#config terminal
```

```
HQ_Router(config)#router ospf 1
```

```
HQ_Router(config-router)#network 10.0.100.0 0.0.0.7 area 0
```

```
HQ_Router(config-router)#network 10.0.200.0 0.0.0.7 area 0
```

```
HQ_Router(config-router)#exit
```

```
HQ_Router(config)#end
```

```
HQ_Router#
```

4.5 HSRP & MLS1 and MLS2 Configuration

HSRP (Hot Standby Router Protocol) is implemented for **network redundancy**.

Working of HSRP

- One router works as the **Active Router**
- Another router works as the **Standby Router**
- If the active router fails, the standby router automatically takes over

This ensures continuous communication without network interruption.

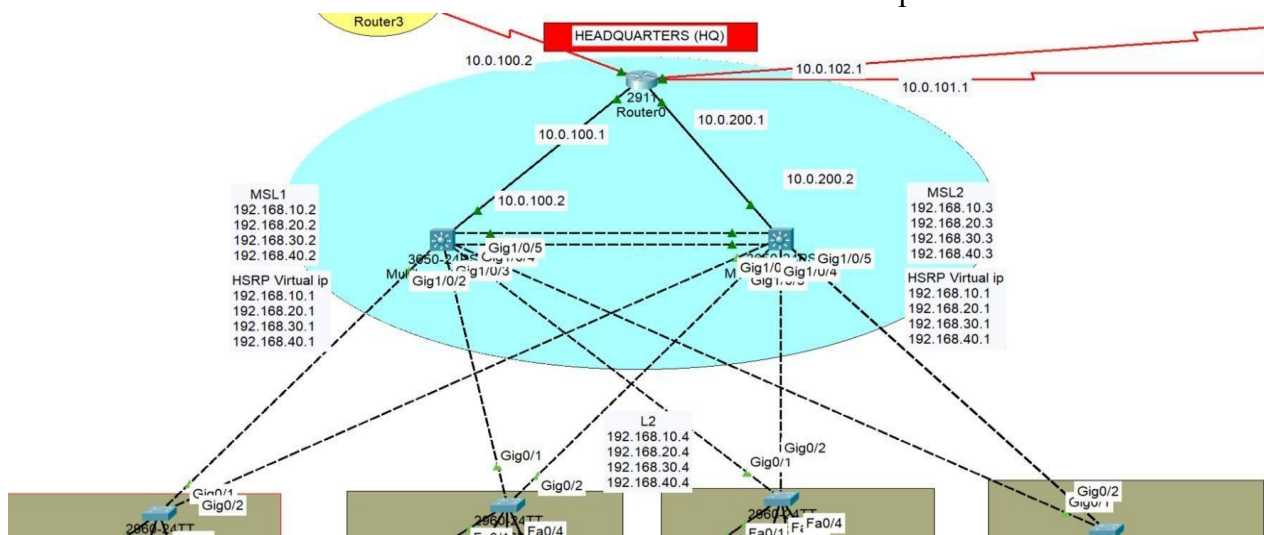


FIGURE 4.5: HSRP Configuration

4.5.1 : HSRP (Hot Standby Router Protocol) Configuration on MSL1

```
MLS1>enable
MLS1#config terminal
```

```
MLS1(config)#interface range gig1/0/2-5
MLS1(config-if-range)#switchport mode trunk
MLS1(config-if-range)#switchport trunk allowed vlan 10,20,30,40
MLS1(config-if-range)#no shutdown
MLS1(config-if-range)#exit
```

```
MLS1(config)#end
MLS1#write
Building configuration...
[OK]
```

```
MLS1>enable
MLS1#config terminal
```

```
MLS1(config)#interface vlan 10
MLS1(config-if)#ip address 192.168.10.2 255.255.255.0
MLS1(config-if)#standby 10 ip 192.168.10.1
MLS1(config-if)#standby 10 priority 110
MLS1(config-if)#standby 10 preempt
MLS1(config-if)#no shutdown
MLS1(config-if)#exit
```

```
MLS1(config)#interface vlan 20
MLS1(config-if)#ip address 192.168.20.2 255.255.255.0
MLS1(config-if)#standby 20 ip 192.168.20.1
MLS1(config-if)#standby 20 priority 100
MLS1(config-if)#standby 20 preempt
MLS1(config-if)#no shutdown
MLS1(config-if)#exit
```

```
MLS1(config)#interface vlan 30
MLS1(config-if)#ip address 192.168.30.2 255.255.255.0
MLS1(config-if)#standby 30 ip 192.168.30.1
MLS1(config-if)#standby 30 priority 110
MLS1(config-if)#standby 30 preempt
MLS1(config-if)#no shutdown
MLS1(config-if)#exit
```

```
MLS1(config)#interface vlan 40
MLS1(config-if)#ip address 192.168.40.2 255.255.255.0
MLS1(config-if)#standby 40 ip 192.168.40.1
```

```
MLS1(config-if)#standby 40 priority 110
MLS1(config-if)#standby 40 preempt
MLS1(config-if)#no shutdown
MLS1(config-if)#exit
```

```
MLS1(config)#do write
Building configuration...
[OK]
```

4.5.2 : EtherChannel (LACP- Link Aggregation Control Protocol) Configuration Between MLS1 and MSL2

```
MLS1>enable
MLS1#configure terminal
```

```
MLS1(config)#interface range gig1/0/6-7
MLS1(config-if-range)#switchport mode trunk
MLS1(config-if-range)#switchport trunk allowed vlan 10,20,30,40
MLS1(config-if-range)#channel-group 1 mode active
MLS1(config-if-range)#no shutdown
MLS1(config-if-range)#exit
```

```
MLS1(config)#interface port-channel 1
MLS1(config-if)#switchport mode trunk
MLS1(config-if)#switchport trunk allowed vlan 10,20,30,40
MLS1(config-if)#no shutdown
MLS1(config-if)#exit
```

```
MLS1(config)#do write
Building configuration...
[OK]
```

Link Configuration Between MLS1 and HQ_Router

```
MLS1>enable
MLS1#config terminal
```

```
MLS1(config)#interface gig1/0/1
MLS1(config-if)#no switchport
MLS1(config-if)#ip address 10.0.100.2 255.255.255.252
MLS1(config-if)#no shutdown
MLS1(config-if)#exit
```

```
MLS1(config)#do write
Building configuration...
[OK]
```

4.5.3 : HSRP (Hot Standby Router Protocol) Configuration on MSL2

```
MSL2>enable
MSL2#config terminal
```

```
MSL2(config)#interface range gig1/0/2-5
MSL2(config-if-range)#switchport trunk allowed vlan remove 1-1005
MSL2(config-if-range)#switchport trunk allowed vlan 10,20,30,40
MSL2(config-if-range)#switchport mode trunk
MSL2(config-if-range)#no shutdown
MSL2(config-if-range)#exit
```

```
MSL2(config)#end
MSL2#write
Building configuration...
[OK]
```

```
MSL2>enable
MSL2#config terminal
```

```
MSL2(config)#interface vlan 10
MSL2(config-if)#ip address 192.168.10.3 255.255.255.0
MSL2(config-if)#standby 10 ip 192.168.10.1
MSL2(config-if)#standby 10 priority 100
MSL2(config-if)#standby 10 preempt
MSL2(config-if)#no shutdown
MSL2(config-if)#exit
```

```
MSL2(config)#interface vlan 20
MSL2(config-if)#ip address 192.168.20.3 255.255.255.0
MSL2(config-if)#standby 20 ip 192.168.20.1
MSL2(config-if)#standby 20 priority 110
MSL2(config-if)#standby 20 preempt
MSL2(config-if)#no shutdown
MSL2(config-if)#exit
```

```
MSL2(config)#interface vlan 30
MSL2(config-if)#ip address 192.168.30.3 255.255.255.0
MSL2(config-if)#standby 30 ip 192.168.30.1
```

```
MSL2(config-if)#standby 30 priority 100
MSL2(config-if)#standby 30 preempt
MSL2(config-if)#no shutdown
MSL2(config-if)#exit
```

```
MSL2(config)#interface vlan 40
MSL2(config-if)#ip address 192.168.40.3 255.255.255.0
MSL2(config-if)#standby 40 ip 192.168.40.1
MSL2(config-if)#standby 40 priority 100
MSL2(config-if)#standby 40 preempt
MSL2(config-if)#no shutdown
MSL2(config-if)#exit
```

```
MSL2(config)#do write
Building configuration...
[OK]
```

4.5.4 : EtherChannel (LACP- Link Aggregation Control Protocol) Configuration Between MLS2 and MLS1 _

```
MSL2>enable
MSL2#configure terminal
```

```
MSL2(config)#interface range gig1/0/6-7
MSL2(config-if-range)#switchport mode trunk
MSL2(config-if-range)#switchport trunk allowed vlan 10,20,30,40
MSL2(config-if-range)#channel-group 1 mode active
MSL2(config-if-range)#no shutdown
MSL2(config-if-range)#exit
```

```
MSL2(config)#interface port-channel 1
MSL2(config-if)#switchport mode trunk
MSL2(config-if)#switchport trunk allowed vlan 10,20,30,40
MSL2(config-if)#no shutdown
MSL2(config-if)#exit
```

```
MSL2(config)#do write
Building configuration...
[OK]
```

4.5.5 : Link Configuration Between MLS2 and HQ_Router

```
MLS1>enable
MLS1#config terminal
```

```

MLS1(config)#interface gig1/0/1
MLS1(config-if)#no switchport
MLS1(config-if)#ip address 10.0.200.2 255.255.255.252
MLS1(config-if)#no shutdown
MLS1(config-if)#exit

```

```

MLS1(config)#do write
Building configuration...
[OK]

```

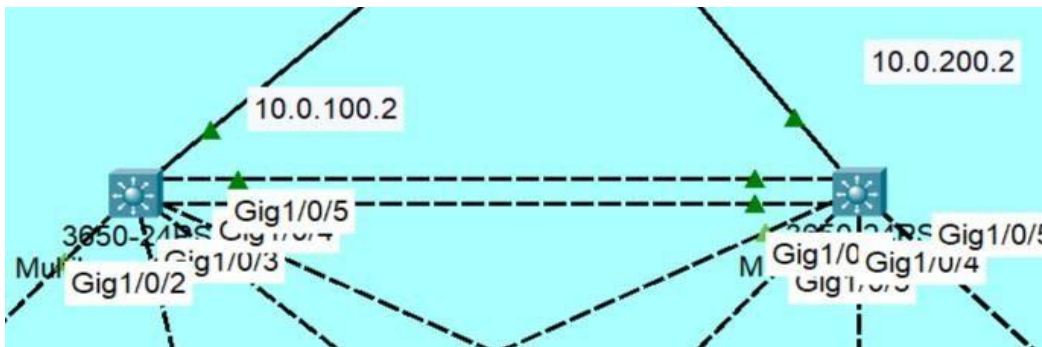
4.6 STP Configuration

STP (Spanning Tree Protocol) is configured to prevent switching loops.

Purpose of STP

- Prevents broadcast storms
- Avoids network loops
- Improves network stability

This helps maintain smooth communication in the network.



4.6.1 : STP (Spanning Tree Protocol) Configuration on MLS1

```

MLS1>enable
MLS1#config terminal

MLS1(config)#spanning-tree vlan 10,20,30,40 priority 4096

MLS1(config)#do write
Building configuration...

```


[OK]

4.6.2 STP (Spanning Tree Protocol) Configuration on MSL1

```
MSL2>enable
```

```
MSL2#config terminal
```

```
MSL2(config)#spanning-tree vlan 10,20,30,40 priority 8192
```

```
MSL2(config)#do write
```

```
Building configuration...
```

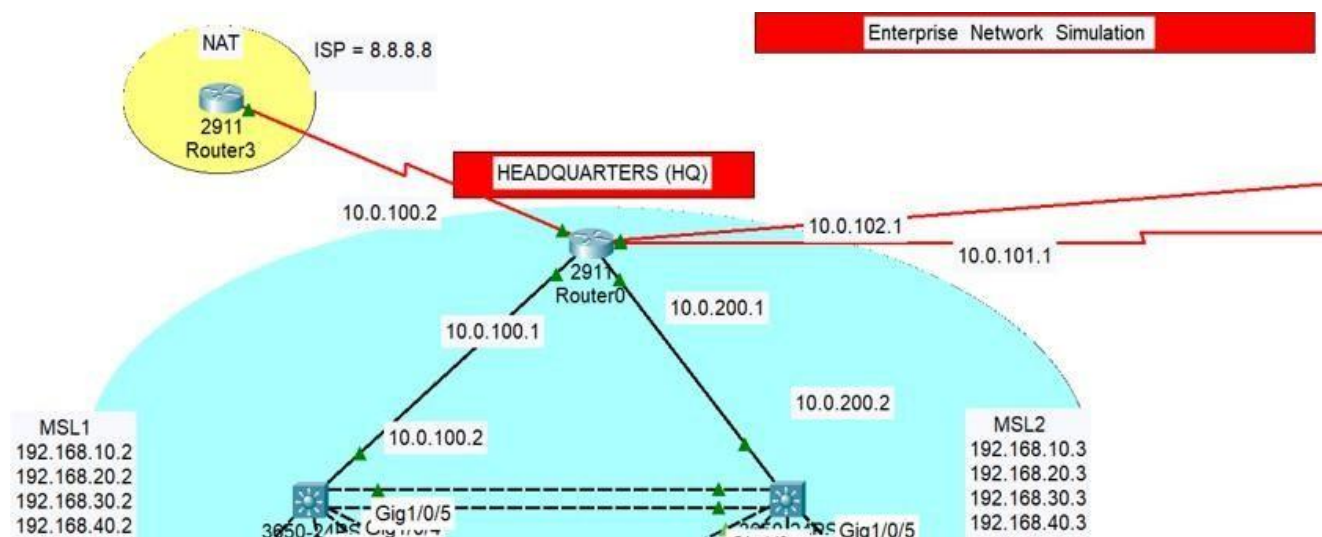
[OK]

4.7 NAT Configuration

NAT (Network Address Translation) is implemented to provide secure internet access.

Purpose of NAT

- Converts private IP addresses to public IP addresses
- Allows internet communication
- Improves network security



This helps devices inside the network access the internet securely.

4.7.1 : ISP Configuration on Isp

```
ISP_Router>enable
ISP_Router#config terminal

ISP_Router(config)#interface s0/0/0
ISP_Router(config-if)#ip address 10.0.150.2 255.255.255.0
ISP_Router(config-if)#no shutdown
ISP_Router(config-if)#exit

ISP_Router(config)#interface loopback 1
ISP_Router(config-if)#ip address 8.8.8.8 255.255.255.255
ISP_Router(config-if)#exit

ISP_Router(config)#ip route 10.0.0.0 255.0.0.0 10.0.150.1
ISP_Router(config)#end
```

4.7.1 : NAT Configuration on HQ_Router

```
HQ_Router>enable
HQ_Router#config terminal

HQ_Router(config)#interface s0/1/0
HQ_Router(config-if)#ip address 10.0.150.1 255.255.255.0
HQ_Router(config-if)#ip nat outside
HQ_Router(config-if)#no shutdown
HQ_Router(config-if)#exit

HQ_Router(config)#interface gig0/0
HQ_Router(config-if)#ip nat inside
HQ_Router(config-if)#exit

HQ_Router(config)#interface gig0/1
HQ_Router(config-if)#ip nat inside
HQ_Router(config-if)#exit

HQ_Router(config)#access-list 1 permit 10.0.100.0 0.0.0.255
HQ_Router(config)#access-list 1 permit 10.0.200.0 0.0.0.255

HQ_Router(config)#ip nat inside source list 1 interface s0/1/0 overload

HQ_Router(config)#ip route 0.0.0.0 0.0.0.0 10.0.150.2
```

```
HQ_Router(config)#end
```

4.7.2 : MLS1 and MLS2 on Default Route

```
MLS1(config)#ip route 0.0.0.0 0.0.0.0 10.0.150.1
```

```
MLS2(config)#ip route 0.0.0.0 0.0.0.0 10.0.150.1
```

4.8 Testing and Verification

After completing the configuration, testing is performed to verify communication.

Testing Performed

- **Ping Test** between HQ and Branch Offices
- **Inter-VLAN Communication Test**
- **OSPF Routing Verification**
- **HSRP Failover Testing**
- **Internet Connectivity Test using NAT**

The test results show successful communication between all departments and branch offices.

4.9 Summary

The implementation phase successfully creates an Enterprise Network Simulation in Cisco Packet Tracer. Technologies such as VLAN, OSPF, HSRP, STP, and NAT are properly configured to provide secure communication, redundancy, routing, and internet access in a real-world enterprise environment.

Chapter 5: Testing & Results

5.1 Introduction

Testing is an important phase of the project to check whether the network is working properly or not. In this project, different tests are performed to verify communication between the **Headquarters (HQ)** and **Branch Offices**, VLAN connectivity, routing, redundancy, and internet access.

The testing process ensures that the enterprise network is working in a **secure, reliable, and efficient manner**.

5.2 Testing Objectives

The main objectives of testing are:

- To check communication between **HQ and Branch Offices**
- To verify **VLAN communication** between departments
- To test **OSPF routing** between networks
- To verify **HSRP failover** during router failure
- To check **STP loop prevention**
- To verify **internet access using NAT**

5.3 Types of Testing Performed

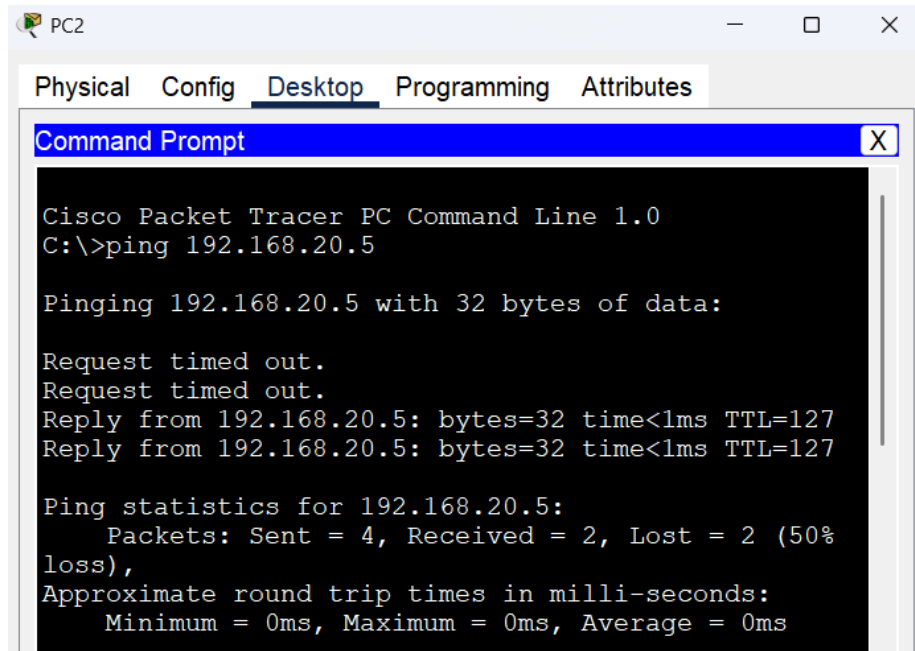
1. Connectivity Testing

This test checks whether devices can communicate with each other.

Example:

- PC from **HQ Manager Department** successfully communicates with **HQ Office PCs**.
- Successful **Ping Test** between offices and VLANs.
- Successful **Ping Test** between Router and Switches.

Result: ✓ Successful



PC2

Physical Config Desktop Programming Attributes

Command Prompt

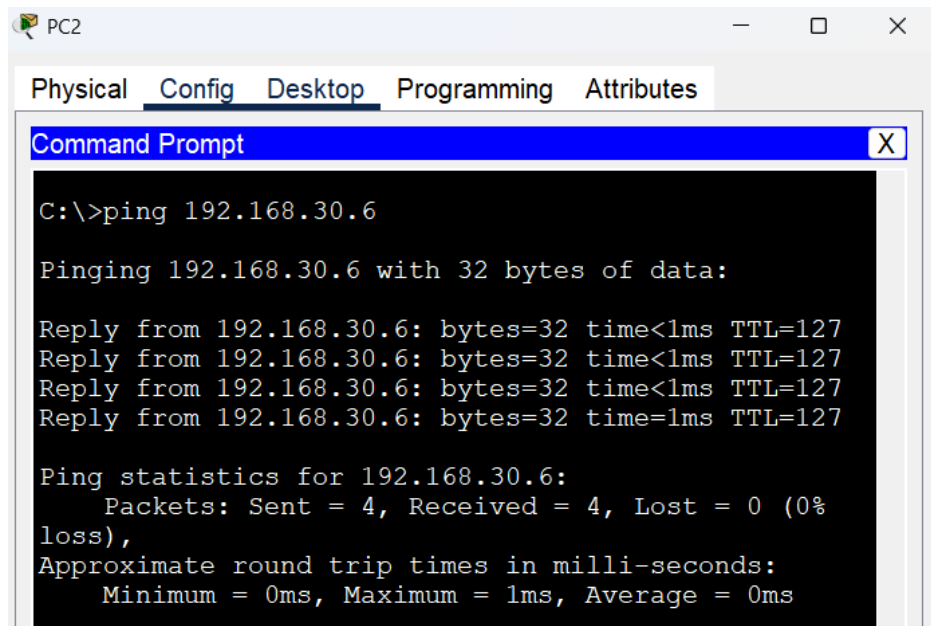
```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.5

Pinging 192.168.20.5 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 192.168.20.5: bytes=32 time<1ms TTL=127
Reply from 192.168.20.5: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.20.5:
    Packets: Sent = 4, Received = 2, Lost = 2 (50%
loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

5.3.1 : Communication Between HQ_Department and IT_Department



PC2

Physical Config Desktop Programming Attributes

Command Prompt

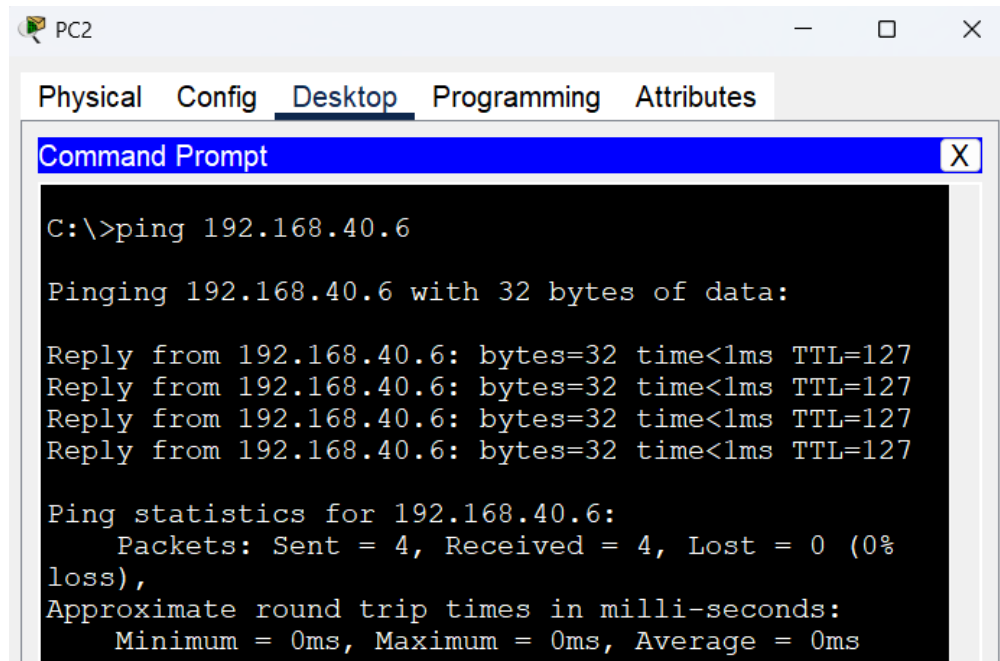
```
C:\>ping 192.168.30.6

Pinging 192.168.30.6 with 32 bytes of data:

Reply from 192.168.30.6: bytes=32 time<1ms TTL=127
Reply from 192.168.30.6: bytes=32 time<1ms TTL=127
Reply from 192.168.30.6: bytes=32 time<1ms TTL=127
Reply from 192.168.30.6: bytes=32 time=1ms TTL=127

Ping statistics for 192.168.30.6:
    Packets: Sent = 4, Received = 4, Lost = 0 (0%
loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

5.3.2 : Communication Between HQ_Department and HR_Department



The screenshot shows a window titled "PC2" with tabs for "Physical", "Config", "Desktop", "Programming", and "Attributes". The "Desktop" tab is active, displaying a "Command Prompt" window. The command prompt shows the execution of the command `C:\>ping 192.168.40.6`. The output indicates a successful ping to 192.168.40.6 with 32 bytes of data. Four replies are shown, each with a time of <1ms and TTL=127. The ping statistics for 192.168.40.6 show 4 packets sent, 4 received, and 0% loss. The approximate round trip times in milli-seconds are: Minimum = 0ms, Maximum = 0ms, Average = 0ms.

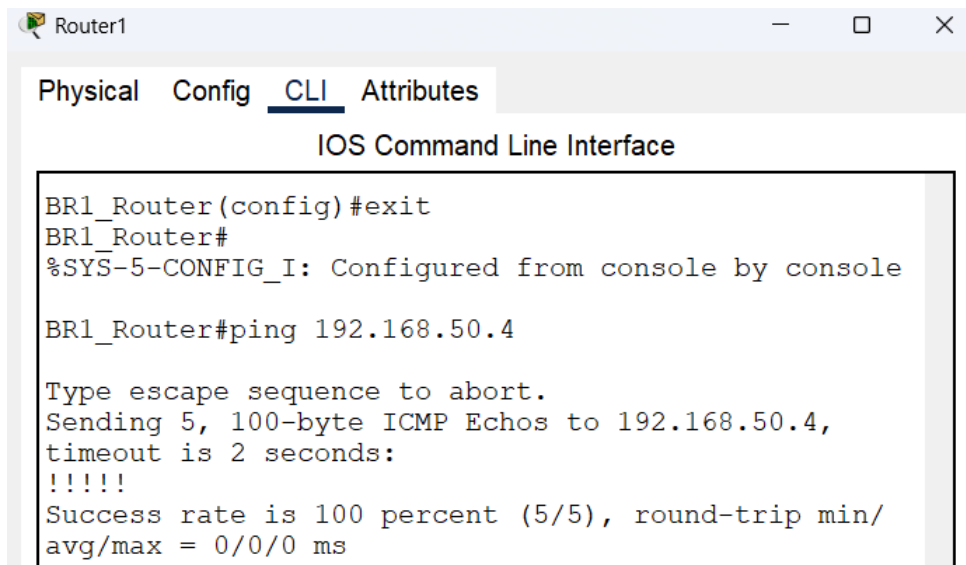
```
C:\>ping 192.168.40.6

Pinging 192.168.40.6 with 32 bytes of data:

Reply from 192.168.40.6: bytes=32 time<1ms TTL=127
Reply from 192.168.40.6: bytes=32 time<1ms TTL=127
Reply from 192.168.40.6: bytes=32 time<1ms TTL=127
Reply from 192.168.40.6: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.40.6:
    Packets: Sent = 4, Received = 4, Lost = 0 (0%
loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

5.3.3 : Communication Between HQ_Department and Sales_Department



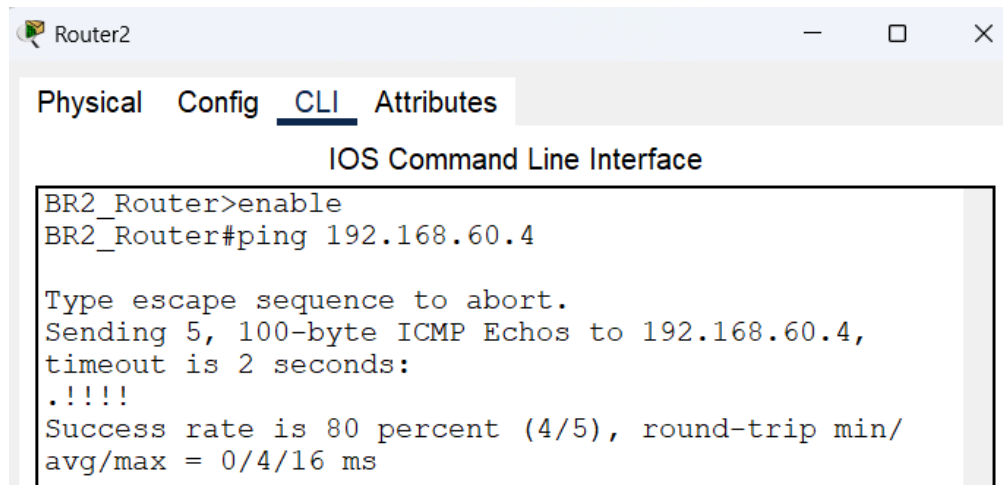
The screenshot shows a window titled "Router1" with tabs for "Physical", "Config", "CLI", and "Attributes". The "CLI" tab is active, displaying the "IOS Command Line Interface". The commands entered are `BR1_Router(config)#exit` and `BR1_Router#`. The output shows the system configuration file being updated: `%SYS-5-CONFIG_I: Configured from console by console`. The command `BR1_Router#ping 192.168.50.4` is entered, and the output shows a successful ping to 192.168.50.4. The output indicates that 5, 100-byte ICMP Echos were sent to 192.168.50.4, with a timeout of 2 seconds. The success rate is 100 percent (5/5), and the round-trip min/avg/max is 0/0/0 ms.

```
BR1_Router(config)#exit
BR1_Router#
%SYS-5-CONFIG_I: Configured from console by console

BR1_Router#ping 192.168.50.4

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.50.4,
timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/
avg/max = 0/0/0 ms
```

5.3.4 : Connection successful Branch 1 Office router to PCs



Router2

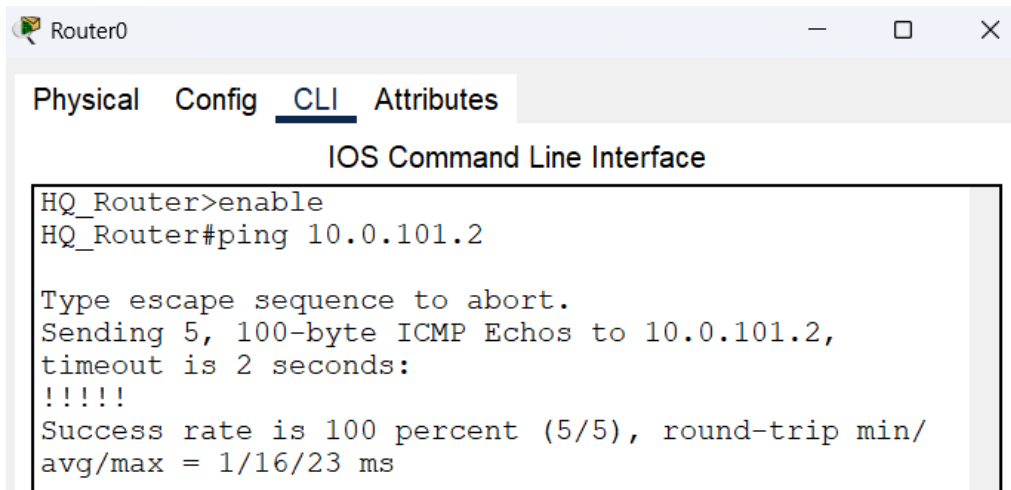
Physical Config CLI Attributes

IOS Command Line Interface

```
BR2_Router>enable
BR2_Router#ping 192.168.60.4

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.60.4,
timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/
avg/max = 0/4/16 ms
```

5.3.5 : Conection successful Branch Office 2 router to Pcs



Router0

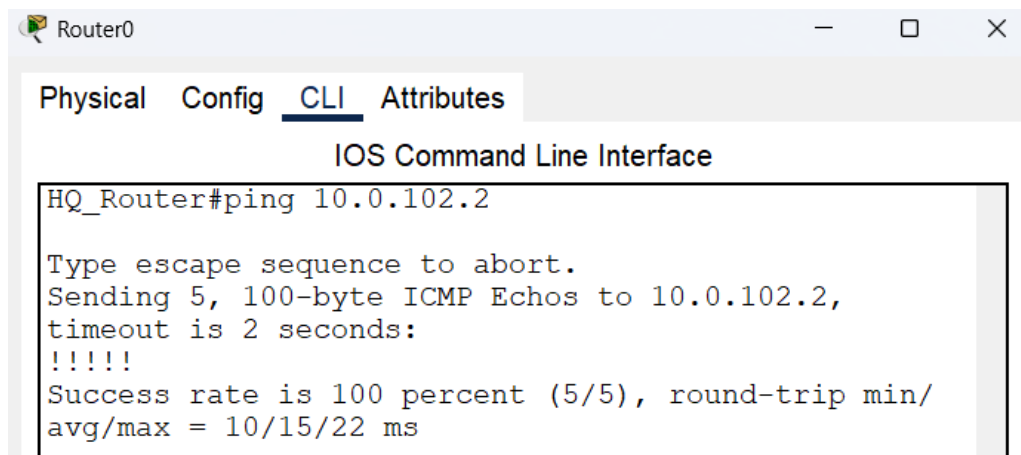
Physical Config CLI Attributes

IOS Command Line Interface

```
HQ_Router>enable
HQ_Router#ping 10.0.101.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.101.2,
timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/
avg/max = 1/16/23 ms
```

5.3.6 : Ping successful HQ router to Branch office 1 Router



Router0

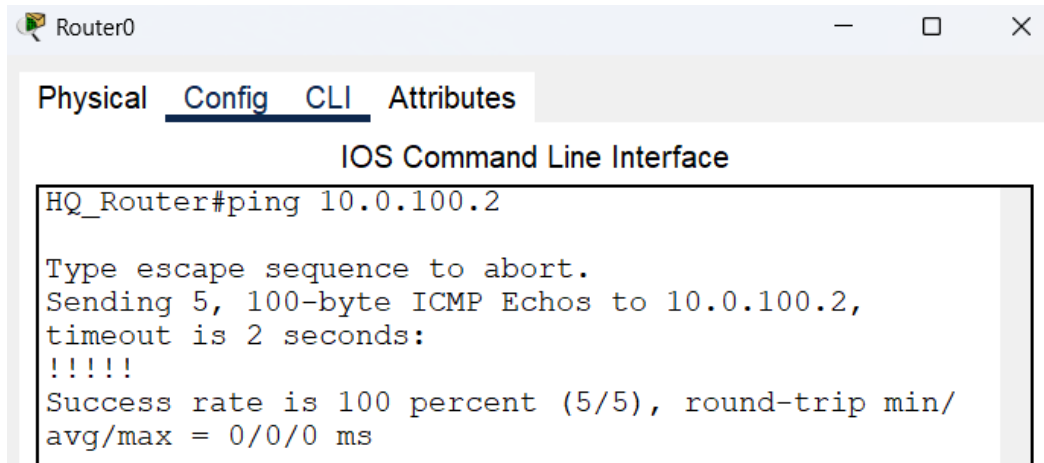
Physical Config CLI Attributes

IOS Command Line Interface

```
HQ_Router#ping 10.0.102.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.102.2,
timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/
avg/max = 10/15/22 ms
```

5.3.7 : Ping successful HQ router to Branch office 2 Router

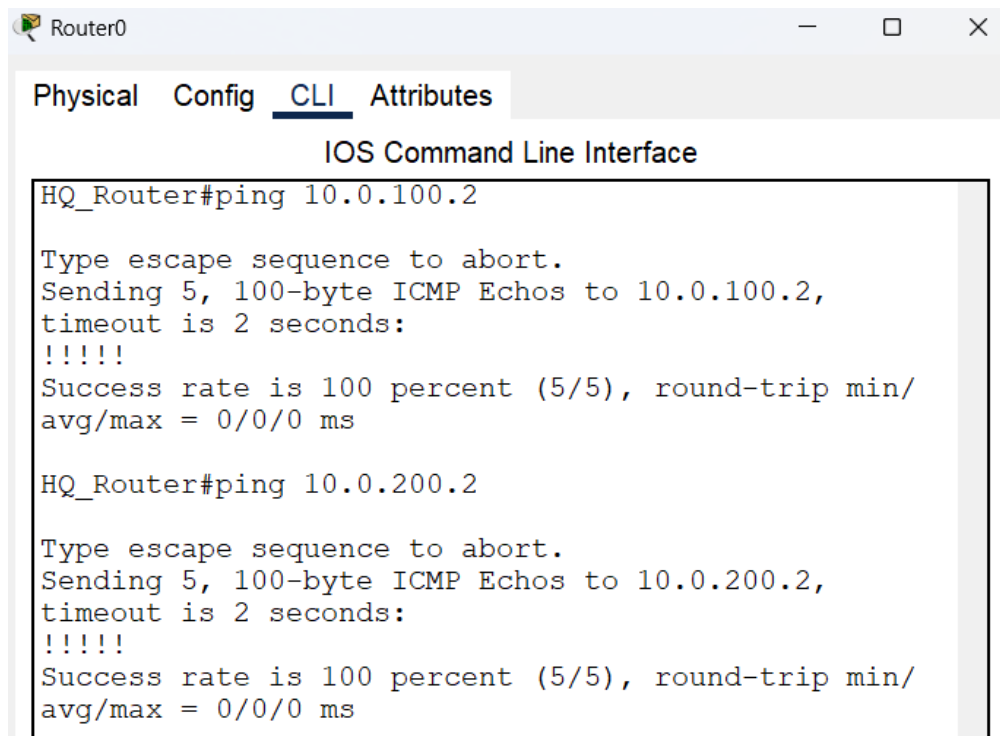


The screenshot shows a window titled "Router0" with tabs for "Physical", "Config", "CLI", and "Attributes". The "CLI" tab is selected, and the "IOS Command Line Interface" is displayed. The command "HQ_Router#ping 10.0.100.2" has been entered, resulting in a successful ping with a 100% success rate.

```
HQ_Router#ping 10.0.100.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.100.2,
timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/
avg/max = 0/0/0 ms
```

5.3.8 : Ping successful HQ router to Isp Router



The screenshot shows the same "Router0" window with the "CLI" tab selected. Two ping commands have been executed successfully: "HQ_Router#ping 10.0.100.2" and "HQ_Router#ping 10.0.200.2". Both resulted in 100% success rates.

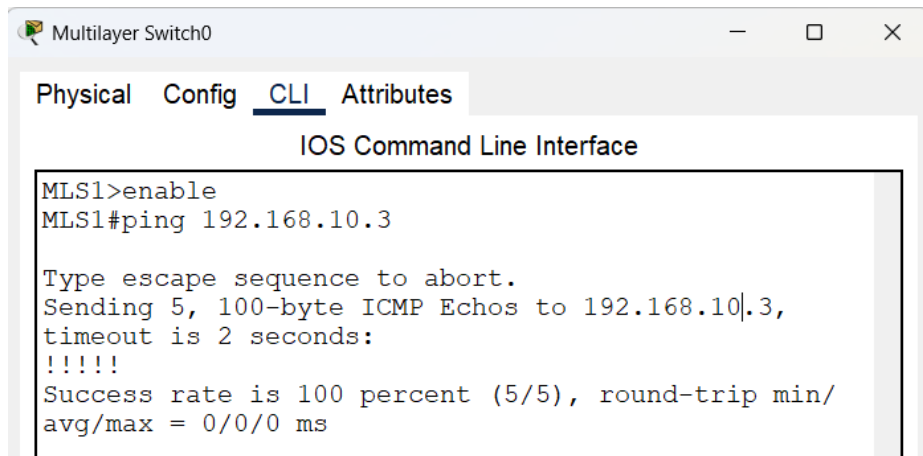
```
HQ_Router#ping 10.0.100.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.100.2,
timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/
avg/max = 0/0/0 ms

HQ_Router#ping 10.0.200.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.200.2,
timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/
avg/max = 0/0/0 ms
```

5.3.9 : Ping successful HQ router to Multi-layer switch 1 &2



Multilayer Switch0

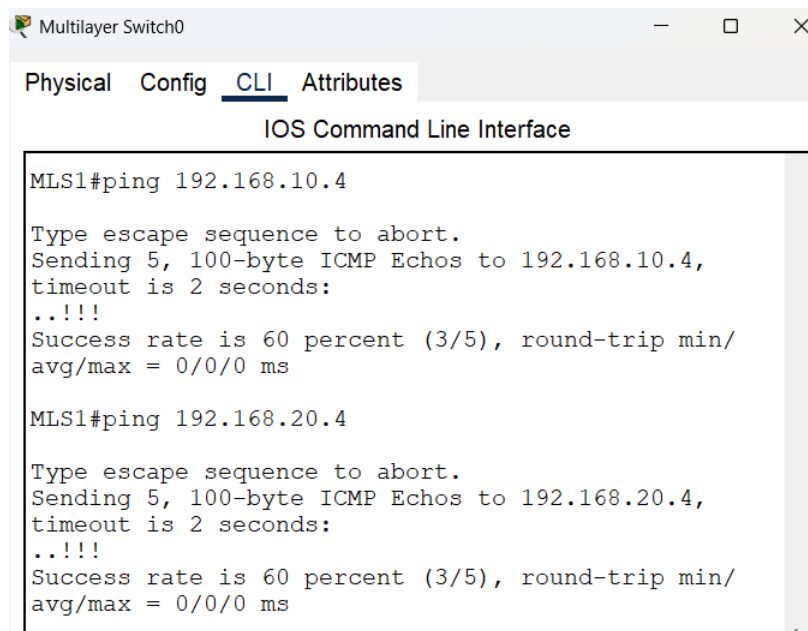
Physical Config CLI Attributes

IOS Command Line Interface

```
MLS1>enable
MLS1#ping 192.168.10.3

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.10.3,
timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/
avg/max = 0/0/0 ms
```

5.3.10 : Ping successfully MLS1 to MLS2



Multilayer Switch0

Physical Config CLI Attributes

IOS Command Line Interface

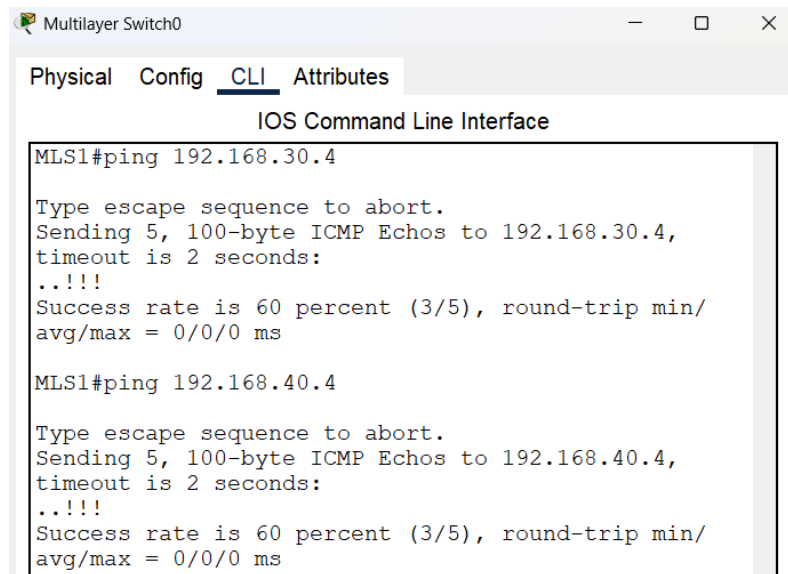
```
MLS1#ping 192.168.10.4

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.10.4,
timeout is 2 seconds:
..!!!
Success rate is 60 percent (3/5), round-trip min/
avg/max = 0/0/0 ms

MLS1#ping 192.168.20.4

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.20.4,
timeout is 2 seconds:
..!!!
Success rate is 60 percent (3/5), round-trip min/
avg/max = 0/0/0 ms
```

5.3.11 : Connection successful MLS1 to vlan 10,20 switches



The screenshot shows a window titled "Multilayer Switch0" with tabs for "Physical", "Config", "CLI", and "Attributes". The "CLI" tab is active, displaying the "IOS Command Line Interface". The terminal output shows two successful ping commands from MLS1 to 192.168.30.4 and 192.168.40.4. Each ping sends 5, 100-byte ICMP Echoes with a 2-second timeout, resulting in a 60 percent success rate (3/5) and 0/0/0 ms round-trip times.

```
Multilayer Switch0
Physical Config CLI Attributes
IOS Command Line Interface
MLS1#ping 192.168.30.4
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echoes to 192.168.30.4,
timeout is 2 seconds:
..!!!
Success rate is 60 percent (3/5), round-trip min/
avg/max = 0/0/0 ms

MLS1#ping 192.168.40.4
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echoes to 192.168.40.4,
timeout is 2 seconds:
..!!!
Success rate is 60 percent (3/5), round-trip min/
avg/max = 0/0/0 ms
```

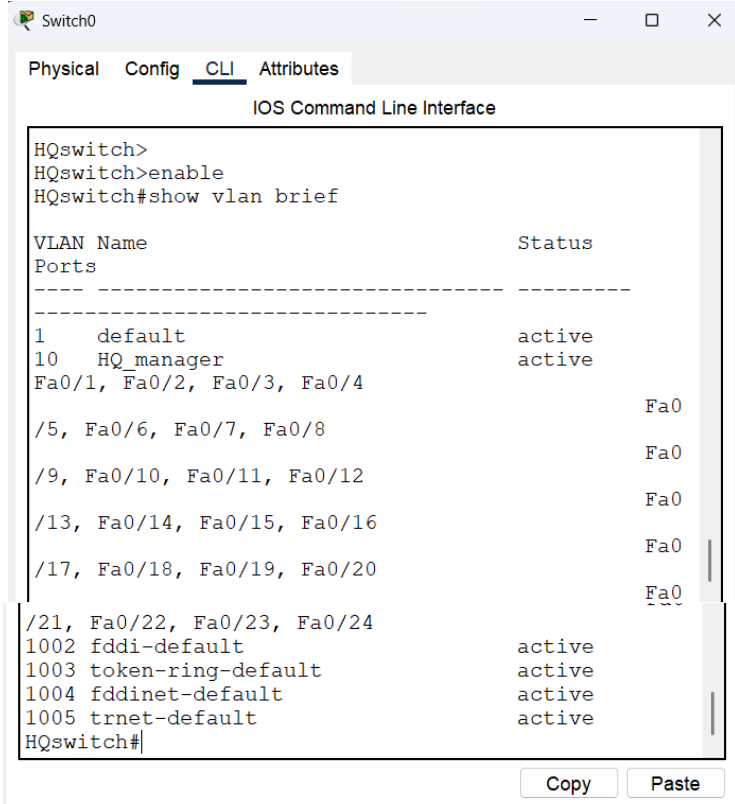
5.3.12 Connection successful MLS1 to vlan 30,40 switches

Result: ✓ Successful

2. VLAN Testing

This test verifies proper communication between different VLANs.

Example:



The screenshot shows a network switch window titled "Switch0" with tabs for Physical, Config, CLI, and Attributes. The CLI tab is active, displaying the "IOS Command Line Interface". The user has entered the following commands:

```
HQswitch>
HQswitch>enable
HQswitch#show vlan brief
```

The output of the "show vlan brief" command is as follows:

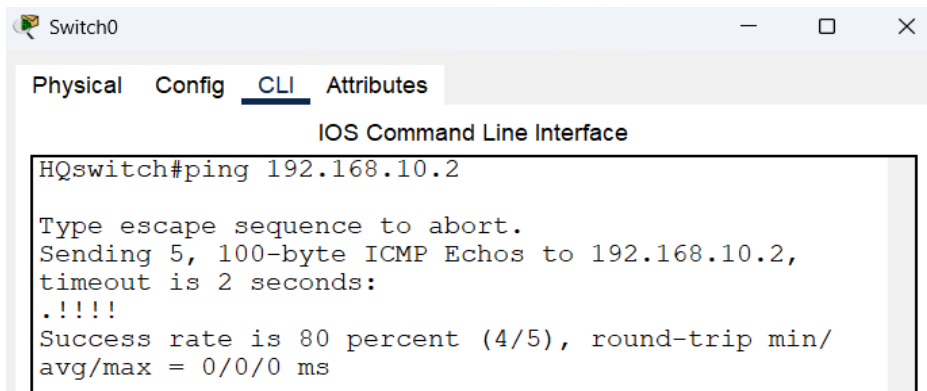
VLAN Name	Status
1 default	active
10 HQ_manager	active
Fa0/1, Fa0/2, Fa0/3, Fa0/4	
/5, Fa0/6, Fa0/7, Fa0/8	
/9, Fa0/10, Fa0/11, Fa0/12	
/13, Fa0/14, Fa0/15, Fa0/16	
/17, Fa0/18, Fa0/19, Fa0/20	
/21, Fa0/22, Fa0/23, Fa0/24	
1002 fddi-default	active
1003 token-ring-default	active
1004 fddinet-default	active
1005 trnet-default	active

The user has returned to the prompt: `HQswitch#`. Below the output, there are "Copy" and "Paste" buttons.

- **VLAN 10 (HQ Manager)** communicates with **VLAN 20 (IT)** through Layer 3 switching.
- Communication between departments works successfully.

VLAN 10 Configuration and Self Testing Successful

Result: ✓ Successful



The screenshot shows the same network switch window "Switch0" with the CLI tab active. The user has entered the following command:

```
HQswitch#ping 192.168.10.2
```

The output of the ping command is as follows:

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.10.2,
timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/
avg/max = 0/0/0 ms
```

Switch1		
Physical	Config	CLI
IOS Command Line Interface		
VLAN Name	Status	
Ports		
-----	-----	
1 default	active	
20 IT_Department	active	
Fa0/1, Fa0/2, Fa0/3, Fa0/4		
	Fa0	
/5, Fa0/6, Fa0/7, Fa0/8		
	Fa0	
/9, Fa0/10, Fa0/11, Fa0/12		
	Fa0	
/13, Fa0/14, Fa0/15, Fa0/16		
	Fa0	
/17, Fa0/18, Fa0/19, Fa0/20		
	Fa0	
/21, Fa0/22, Fa0/23, Fa0/24		
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

Switch1		
Physical	Config	CLI
IOS Command Line Interface		
IT_Department>ping 192.168.20.2		
Type escape sequence to abort.		
Sending 5, 100-byte ICMP Echos to 192.168.20.2,		
timeout is 2 seconds:		
.!!!!		
Success rate is 80 percent (4/5), round-trip min/		
avg/max = 0/0/0 ms		

VLAN 20 Configuration and Self Testing Sucessful

Result: ✓ Successful

Switch2

Physical
Config
CLI
Attributes

IOS Command Line Interface

VLAN Name	Status
Ports	

1 default	active
30 HR_Department	active
Fa0/1, Fa0/2, Fa0/3, Fa0/4	
	Fa0
/5, Fa0/6, Fa0/7, Fa0/8	
	Fa0
/9, Fa0/10, Fa0/11, Fa0/12	
	Fa0
/13, Fa0/14, Fa0/15, Fa0/16	
	Fa0
/17, Fa0/18, Fa0/19, Fa0/20	
	Fa0
/21, Fa0/22, Fa0/23, Fa0/24	
1002 fddi-default	active
1003 token-ring-default	active
1004 fddinet-default	active
1005 trnet-default	active

```

HR_Department#ping 192.168.30.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.30.2,
timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/
avg/max = 0/0/0 ms

```

VLAN 30 Configuration and Self Testing Successful

Switch3

Physical
Config
CLI
Attributes

IOS Command Line Interface

VLAN Name	Status
Ports	

1 default	active
40 Sales_Department	active
Fa0/1, Fa0/2, Fa0/3, Fa0/4	
	Fa0
/5, Fa0/6, Fa0/7, Fa0/8	
	Fa0
/9, Fa0/10, Fa0/11, Fa0/12	
	Fa0
/13, Fa0/14, Fa0/15, Fa0/16	
	Fa0
/17, Fa0/18, Fa0/19, Fa0/20	
	Fa0
/21, Fa0/22, Fa0/23, Fa0/24	
1002 fddi-default	active
1003 token-ring-default	active
1004 fddinet-default	active
1005 trnet-default	active

```

Sales_department#ping 192.168.40.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.40.2,
timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/
avg/max = 0/0/0 ms

```

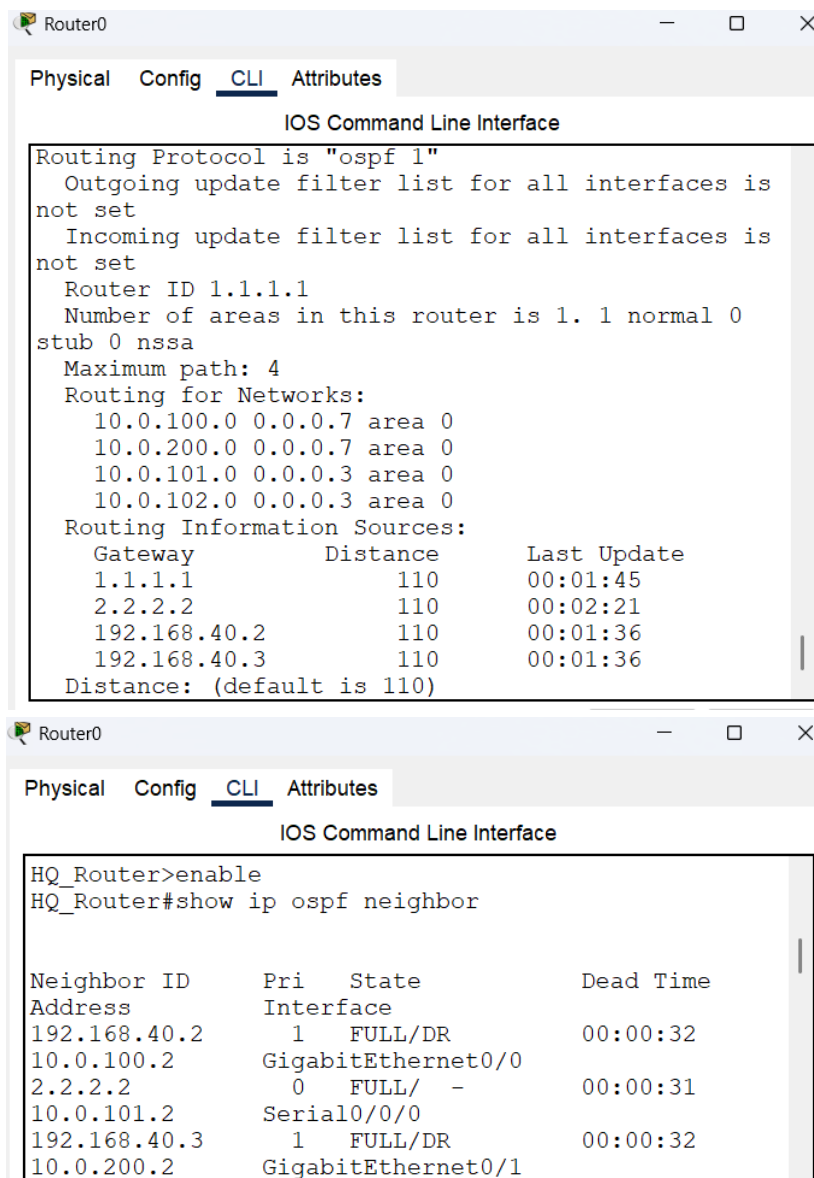
VLAN 40 Configuration and Self Testing Successful

3. OSPF Routing Testing

This test verifies dynamic routing between HQ and branch offices.

Example:

- Routes are automatically learned through **OSPF (Open Shortest Path First)**.
- Branch Office 1 and Branch Office 2 successfully communicate with HQ.



The image shows two screenshots of a Cisco IOS CLI interface. The top screenshot displays the output of the 'show ip ospf' command on Router0, showing OSPF configuration details. The bottom screenshot shows the output of the 'show ip ospf neighbor' command on HQ_Router, displaying a table of OSPF neighbors.

Router0 Output:

```
Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is
not set
  Incoming update filter list for all interfaces is
not set
  Router ID 1.1.1.1
  Number of areas in this router is 1. 1 normal 0
stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    10.0.100.0 0.0.0.7 area 0
    10.0.200.0 0.0.0.7 area 0
    10.0.101.0 0.0.0.3 area 0
    10.0.102.0 0.0.0.3 area 0
  Routing Information Sources:
    Gateway         Distance      Last Update
    1.1.1.1          110          00:01:45
    2.2.2.2          110          00:02:21
    192.168.40.2     110          00:01:36
    192.168.40.3     110          00:01:36
  Distance: (default is 110)
```

HQ_Router Output:

```
HQ_Router>enable
HQ_Router#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time
192.168.40.2	1	FULL/DR	00:00:32
10.0.100.2		GigabitEthernet0/0	
2.2.2.2	0	FULL/ -	00:00:31
10.0.101.2		Serial0/0/0	
192.168.40.3	1	FULL/DR	00:00:32
10.0.200.2		GigabitEthernet0/1	

Result: ✓ Successful

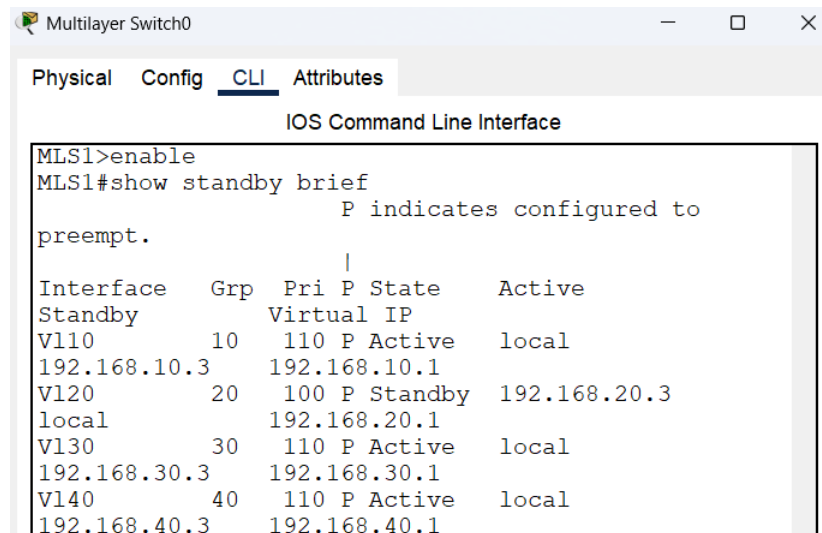
Check OSPF Protocols & OSPF Neighbours

4. HSRP Failover Testing

This test checks backup router functionality.

Example:

- When the main router is turned off, the standby router automatically takes over.



```
Multilayer Switch0
Physical Config CLI Attributes
IOS Command Line Interface
MLS1>enable
MLS1#show standby brief
P indicates configured to preempt.
Interface  Grp  Pri P State      Active
Standby    Virtual IP
Vl10       10   110 P Active    local
192.168.10.3 192.168.10.1
Vl20       20   100 P Standby   192.168.20.3
local      192.168.20.1
Vl30       30   110 P Active    local
192.168.30.3 192.168.30.1
Vl40       40   110 P Active    local
192.168.40.3 192.168.40.1
```

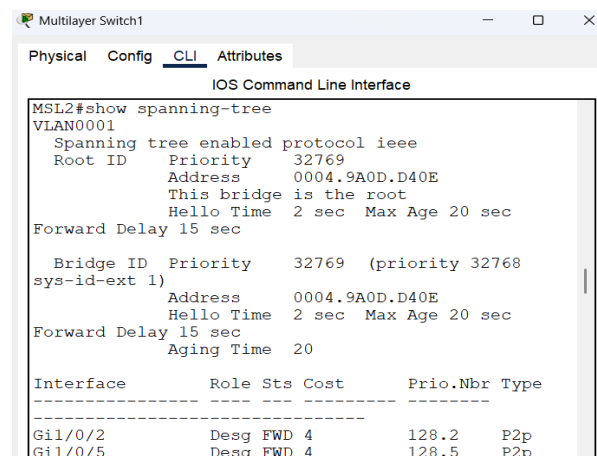
Result: ✓ Successful

5. STP Testing

This test verifies loop prevention in the switching network.

Example:

- Redundant links are managed without creating network loops.



```
Multilayer Switch1
Physical Config CLI Attributes
IOS Command Line Interface
MSL2#show spanning-tree
VLAN0001
Spanning tree enabled protocol ieee
Root ID    Priority    32769
Address    0004.9A0D.D40E
This bridge is the root
Hello Time 2 sec Max Age 20 sec
Forward Delay 15 sec
Bridge ID  Priority    32769 (priority 32768
sys-id-ext 1)
Address    0004.9A0D.D40E
Hello Time 2 sec Max Age 20 sec
Forward Delay 15 sec
Aging Time 20
Interface  Role Sts Cost      Prio.Nbr Type
-----
Gi1/0/2    Desg FWD 4         128.2    P2p
Gi1/0/5    Desg FWD 4         128.5    P2p
```

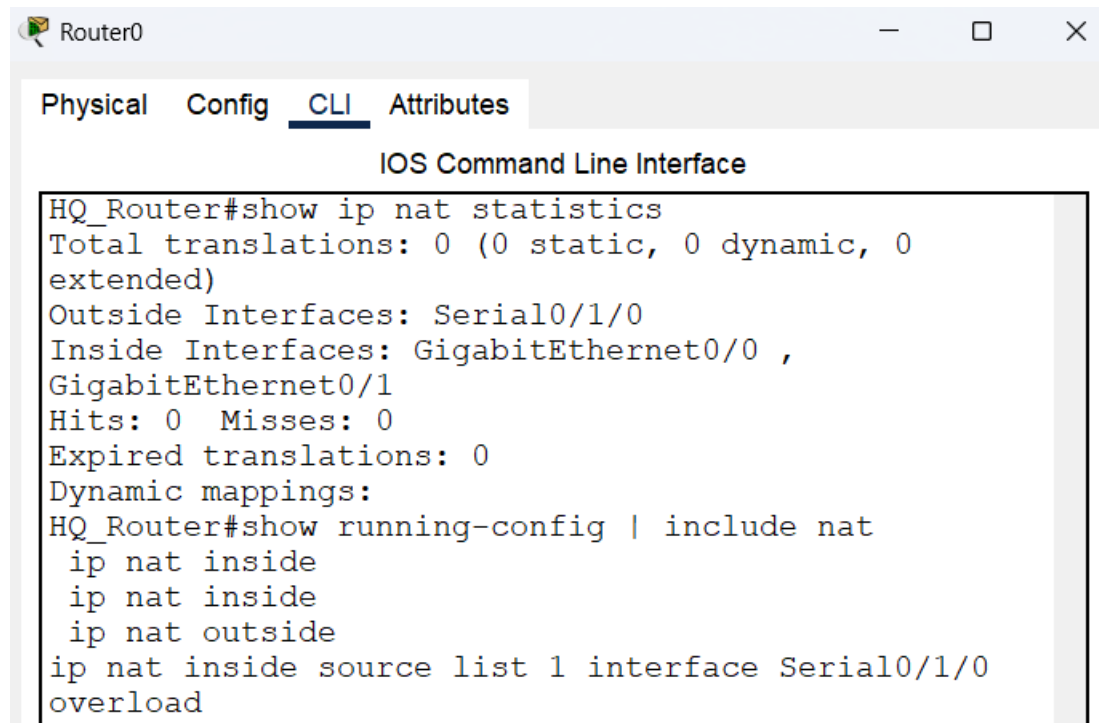
Result: ✓ Successful

6. NAT Testing

This test verifies internet access through NAT.

Example:

- Internal private IP addresses successfully access the internet.



```
Router0
Physical Config CLI Attributes
IOS Command Line Interface
HQ_Router#show ip nat statistics
Total translations: 0 (0 static, 0 dynamic, 0
extended)
Outside Interfaces: Serial0/1/0
Inside Interfaces: GigabitEthernet0/0 ,
GigabitEthernet0/1
Hits: 0 Misses: 0
Expired translations: 0
Dynamic mappings:
HQ_Router#show running-config | include nat
ip nat inside
ip nat inside
ip nat outside
ip nat inside source list 1 interface Serial0/1/0
overload
```

Result: ✓ Successful

5.4 Result Analysis

After testing, it is observed that the network works successfully and provides smooth communication between all departments and branch offices.

Final Results

- Secure communication achieved ✓
- HQ and Branch Offices connected successfully ✓
- VLAN segmentation working properly ✓
- OSPF routing functioning correctly ✓
- HSRP backup working successfully ✓
- STP preventing loops ✓

- NAT providing secure internet access ✓

5.5 Summary

The testing results confirm that the **Enterprise Network Simulation using Cisco Packet Tracer** works successfully. All networking technologies such as **VLAN, OSPF, HSRP, STP, and NAT** are functioning properly and provide a **secure, scalable, and reliable enterprise network environment**.

S. No.	Testing Type	Purpose	Result
1	Connectivity Testing	To verify communication between HQ and Branch Offices	✓ Successful
2	VLAN Testing	To check communication between different VLANs/departments	✓ Successful
3	OSPF Routing Testing	To verify dynamic routing and route exchange between routers	✓ Successful
4	HSRP Failover Testing	To check backup router functionality during failure	✓ Successful
5	STP Testing	To prevent switching loops and broadcast storms	✓ Successful
6	NAT Testing	To verify internet access through private-to-public IP translation	✓ Successful

Chapter 6: Conclusion & Future Scope

6.1 Conclusion

The project “**Enterprise Network Simulation using Cisco Packet Tracer**” was successfully designed, implemented, and tested to create a secure, reliable, and scalable enterprise network environment. The main purpose of this project was to understand how a real company network works and how communication is maintained between different departments and branch offices. In this project, a **Headquarters (HQ)** and **two Branch Offices** were connected using different networking technologies to simulate a real-life enterprise network system. The project helped in understanding how organizations manage communication, routing, security, redundancy, and internet access efficiently.

In today’s digital world, every company depends on a secure and fast network system to perform daily business activities. Large organizations contain many departments such as **Management, Human Resources (HR), Information Technology (IT), and Sales**, and these departments need continuous communication to share files, employee records, business reports, customer information, and technical resources. If all departments are connected in one single network without proper management, it may create problems such as **high network traffic, poor communication speed, security issues, and network failure**. Therefore, designing a structured and secure enterprise network becomes very important.

This project was developed to solve such networking problems by implementing important networking technologies such as **VLAN (Virtual Local Area Network), OSPF (Open Shortest Path First), HSRP (Hot Standby Router Protocol), STP (Spanning Tree Protocol), and NAT (Network Address Translation)**. These technologies helped in improving network performance, security, reliability, communication, and internet access.

The first major achievement of this project was the successful implementation of **VLAN (Virtual Local Area Network)**. VLAN technology was used to divide the network into separate departments such as **HQ Manager Department, IT Department, HR Department, and Sales Department**. By using VLANs, each department was separated logically, which reduced unnecessary network traffic and improved communication speed. It also increased security because devices from one department were controlled separately from other departments. This helped in creating a more organized and manageable network structure.

Another important achievement of this project was the successful implementation of **OSPF (Open Shortest Path First)** routing protocol. Since the enterprise network consisted of a Headquarters and multiple Branch Offices, proper communication between offices was very important. OSPF was configured to provide dynamic routing between different routers and office locations. One major advantage of OSPF is that it automatically finds the best path for communication. If one network path becomes unavailable, OSPF automatically updates the routing table and selects another path for communication. This feature improved communication reliability and ensured smooth connectivity between HQ and branch offices.

The project also successfully implemented **HSRP (Hot Standby Router Protocol)** to provide network redundancy and backup connectivity. In real organizations, if the main router fails, the entire communication system may stop working, causing major business problems. To solve this issue, HSRP was configured in the network where one router worked as an **active router** and another worked as a **standby router**. During testing, when the active router was turned off, the standby router automatically took control and maintained communication without interruption. This proved that the network can continue functioning even during hardware failure.

Another important technology implemented in this project was **STP (Spanning Tree Protocol)**. In enterprise networks, multiple switches are connected to improve network communication and backup connectivity. However, multiple switch connections may create loops, which can crash the network or create broadcast storms. STP was used to solve this problem by blocking unnecessary paths and selecting the best route for communication. This improved network stability and prevented switching loops successfully.

The project also implemented **NAT (Network Address Translation)** to provide secure internet communication. Most internal networks use private IP addresses that cannot directly access the internet. NAT was configured to convert private IP addresses into public IP addresses for internet communication. After successful configuration, devices inside the network were able to communicate externally, proving that internet access was successfully achieved in the enterprise network environment.

The network was successfully designed and implemented using **Cisco Packet Tracer**, which is a simulation software widely used for networking practice and education. This software helped in creating a practical networking environment without requiring expensive hardware devices. Various networking devices such as **routers, switches, PCs, and servers** were configured to simulate a real company environment. Cisco Packet Tracer also helped in testing communication between departments and branch offices effectively.

Different tests were performed to verify the proper functioning of the network. These tests included:

1. Connectivity Testing

Connectivity testing was performed to verify communication between different offices and departments. Ping tests were conducted between Headquarters and Branch Offices to ensure successful communication. The results showed successful connectivity across the network.

2. VLAN Communication Testing

The communication between different VLANs was tested to ensure proper department interaction. The results confirmed successful communication between Manager, IT, HR, and Sales departments through Layer 3 switching.

3. OSPF Routing Testing

OSPF routing verification was performed to check route exchange between routers. It was observed that all routers successfully learned routes dynamically and maintained smooth communication.

4. HSRP Failover Testing

The failover mechanism of HSRP was tested by disabling the active router. The standby router automatically became active and maintained communication successfully.

5. STP Testing

STP functionality was tested to verify loop prevention. Redundant links were successfully managed without causing network failure.

6. NAT Testing

NAT was tested to verify internet access. Internal private IP addresses were successfully translated and communication with external networks was achieved.

The testing phase proved that the project worked successfully and fulfilled all project objectives. The enterprise network achieved **secure communication, reliable routing, redundancy, department separation, and internet access** successfully.

This project also provided valuable practical learning experience in networking technologies. Through this project, practical knowledge of **router configuration, switch management, VLAN creation, routing protocols, network redundancy, and troubleshooting** was gained. It also improved understanding of how enterprise organizations maintain secure communication between different offices and departments.

Another important benefit of this project is its usefulness in future career opportunities. Knowledge gained from this project can be helpful in careers related to **Networking, System Administration, Cybersecurity, Cloud Computing, and IT Infrastructure Management**. Since many companies use Cisco networking technologies, this project also improves practical understanding required for networking certifications such as CCNA.

Overall, this project successfully demonstrates how a secure, scalable, and reliable enterprise network can be created using Cisco Packet Tracer. The implementation of VLAN, OSPF, HSRP, STP, and NAT technologies helped in building a strong communication system between Headquarters and Branch Offices. Therefore, it can be concluded that the project achieved all its objectives successfully and provided practical knowledge of enterprise networking in a real-world environment.

6.2 Future Scope

Although the project was successfully implemented, many advanced features and technologies can be added in the future to improve network performance, scalability, and security.

6.2.1. Adding More Branch Offices

In the current project, only two branch offices are connected with Headquarters. In the future, more branch offices can be added in different locations to simulate a larger enterprise environment. This will improve understanding of large-scale network communication.

For example, suppose a company has its **Headquarters (HQ) in Delhi**, and currently two branch offices are located in **Noida and Gurgaon**. In the future, additional branch offices can be added in cities such as **Mumbai, Patna, Bengaluru, Kolkata, and Hyderabad**. All these branch offices can be connected with the HQ through routing protocols like **OSPF (Open Shortest Path First)** for smooth communication.

For instance:

- **HQ Office → Delhi**
- **Branch Office 1 → Noida**
- **Branch Office 2 → Gurgaon**
- **Branch Office 3 → Mumbai**
- **Branch Office 4 → Patna**
- **Branch Office 5 → Bengaluru**

In this setup, employees working in different cities will be able to communicate with each other, share important company files, access common servers, and exchange business information securely. For example, if the **HR Department in Delhi HQ** wants to send employee details to the **Patna Branch Office**, the communication can happen quickly through the enterprise network.

Similarly, if the **Sales Department in Mumbai** wants to access customer records stored at the **HQ server**, they can do it without any delay. This will improve business communication and productivity.

Adding more branch offices will also help in learning **large-scale network management**, routing configuration, traffic handling, and network scalability. It will make the project closer to a real enterprise network used by large organizations such as banks, hospitals, IT companies, and multinational companies.

Therefore, future expansion by adding more branch offices can improve the network's **coverage, communication speed, reliability, scalability, and real-world implementation capability**.

6.2.2. Advanced Security Features

Future improvements may include advanced security mechanisms such as:

- Firewall configuration
- Access Control Lists (ACL)
- Intrusion Detection System (IDS)
- Intrusion Prevention System (IPS)
- VPN (Virtual Private Network)

These technologies will help protect sensitive company data from unauthorized access and cyber attacks.

6.23. Wireless Networking

Currently, the project uses wired communication only. In the future, **wireless networking (Wi-Fi)** can be implemented to provide mobile connectivity for employees, laptops, and smart devices.

For example, suppose the **Headquarters (HQ)** of a company has different departments such as **HR, IT, Sales, and Management**. Currently, employees use desktop computers connected through LAN cables. In the future, wireless access points can be installed in different departments so that employees can connect using laptops and mobile devices.

For instance:

- **HR Department** employees can use laptops to access employee records through Wi-Fi.
- **Sales Department** employees can attend online meetings and update customer information from mobile devices.
- **Managers** can move freely within the office while staying connected to company emails and reports.
- **Visitors or guests** can be provided with a separate guest Wi-Fi network for internet access.

Similarly, in **Branch Offices**, employees can access the company server wirelessly without depending completely on cables. For example, an employee in **Branch Office 1** can connect a laptop through Wi-Fi and access files stored in the **HQ office server** securely.

Wireless networking will also support modern office technologies such as:

- **Video conferencing**
- **Online meetings**
- **Cloud applications**
- **Smart office devices**
- **Remote work support**

However, proper security measures such as **Wi-Fi passwords, WPA2/WPA3 encryption, firewall protection, and access control** should be implemented to prevent unauthorized access to the network.

Therefore, adding wireless networking in the future will improve **flexibility, employee mobility, communication efficiency, and modern workplace connectivity**, making the enterprise network more suitable for real-world business environments.

6.24. Cloud Integration

Cloud computing is widely used in modern organizations. Future enhancement can include cloud integration for:

- Remote file access
- Online storage
- Cloud applications
- Backup services

This will make the network more flexible and modern.

6.2.5. Network Monitoring System

Network monitoring tools can be implemented to continuously monitor network performance and troubleshoot issues. Monitoring tools help identify failures quickly and improve system management.

Network monitoring tools continuously check:

- Network traffic performance
- Router and switch status
- Internet connectivity
- Packet transmission
- Network failures and security threats
- Device communication between HQ and Branch Offices

One commonly used network monitoring tool is **Wireshark**.

6.2.6. Real Hardware Implementation

Currently, the network is simulated in Cisco Packet Tracer. In the future, the same project can be implemented using **real Cisco routers, switches, and networking devices**. This will provide hands-on industry-level experience.

6.2.7. Better Routing Technologies

More advanced routing protocols such as **EIGRP** or **BGP** can be implemented to improve communication in large enterprise environments.

6.2.8. IPv6 Implementation

Currently, the network uses IPv4 addressing. Since IPv4 addresses are limited, future implementation may include **IPv6 addressing**, which provides a larger address space and better communication features.

6.2.9. Cybersecurity Enhancements

Future improvements may also include:

- Network encryption
- Secure authentication systems
- Endpoint security
- Multi-factor authentication

These technologies will improve overall network safety.

Future Scope Summary

In the future, this project can become more advanced by adding better security, cloud technology, wireless networking, real hardware devices, advanced routing, and cybersecurity features. These improvements will make the network more suitable for real-world enterprise environments and modern business requirements.

REFERENCES / BIBLIOGRAPHY

The following books, websites, research materials, and official documentation were referred to during the development of the project “**Enterprise Network Simulation using Cisco Packet Tracer**”. These references helped in understanding networking concepts, routing protocols, VLAN implementation, redundancy, switching, and internet connectivity in enterprise environments.

Books References

1. Computer Networking: A Top-Down Approach
Kurose, James F., and Keith W. Ross. *Computer Networking: A Top-Down Approach*. Pearson Education.
This book was used to understand networking fundamentals, routing concepts, protocols, and enterprise communication systems.
2. Data Communications and Networking
Forouzan, Behrouz A. *Data Communications and Networking*. McGraw-Hill Education.
This reference helped in understanding IP addressing, routing, switching, and communication protocols.
3. CCNA Routing and Switching Study Guide
Lammle, Todd. *CCNA Routing and Switching Study Guide*. Sybex Publications.
This book was used for practical understanding of VLAN, OSPF, NAT, HSRP, and STP configurations.
4. Computer Networks
Tanenbaum, Andrew S. *Computer Networks*. Pearson Education.
This book provided theoretical knowledge of network architecture and communication systems.

Online Resources and Official Documentation

5. Cisco Systems Official Documentation
The official Cisco documentation was referred to for understanding router and switch configurations, network troubleshooting, and enterprise technologies.
[Cisco Official Website](#)
6. [Cisco Networking Academy \(NetAcad\)](#)
Cisco Networking Academy learning materials were used to understand enterprise networking, routing, and switching practically through Cisco Packet Tracer.
7. [GeeksforGeeks Computer Networking Tutorials](#)
This website helped in understanding networking concepts such as VLAN, OSPF, HSRP, NAT, and STP with practical examples.

8. **TutorialsPoint Networking Guide**
TutorialsPoint was used as an additional learning source for networking technologies and protocol explanations.
9. [Cisco Packet Tracer Download and Information](#)
Cisco Packet Tracer software and related study material were used to simulate the enterprise network environment.

Research and Learning Materials

10. Department laboratory manuals, practical records, networking notes, and faculty guidance from the Department of Computer Science & Engineering, Dr. C.V. Raman University, Vaishali (Bihar).
11. Internet-based study materials, networking tutorials, and online educational resources related to enterprise networking and Cisco technologies were also referred to during the project development process.

Summary

The above references provided theoretical and practical knowledge required for designing and implementing the **Enterprise Network Simulation using Cisco Packet Tracer** project. These resources helped in understanding enterprise network design, routing protocols, switching technologies, redundancy, and secure internet communication.

Appendices

Appendix A: Network Topology Diagram

This appendix contains the **Enterprise Network Topology Diagram** created in **Cisco Packet Tracer**, showing the connection between **Headquarters (HQ)**, **Branch Office 1**, and **Branch Office 2** along with routers, switches, PCs, and servers.

Appendix B: IP Addressing Scheme

Department / Office	VLAN ID	IP Address Range
HQ Manager Department	10	192.168.10.5 to 192.168.10.255/24
IT Department	20	192.168.20.5 to 192.168.20.255/24
HR Department	30	192.168.30.5 to 192.168.30.255/24
Sales Department	40	192.168.40.5 to 192.168.40.255/24
Branch Office 1	50	192.168.50.3 to 192.168.50.255/24
Branch Office 2	60	192.168.60.3 to 192.168.60.255/24

Appendix C: Network Devices Used

The following networking devices are used in this project:

- **Cisco 2911 Routers**



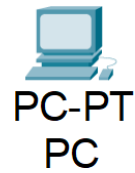
- **Layer 3 (Multilayer) Switches**



- **Layer 2 Switches**



- **PCs (Personal Computers)**



- **Servers**



- **Printers**



Appendix D: Technologies Used

- **VLAN (Virtual Local Area Network)**
- **OSPF (Open Shortest Path First)**
- **HSRP (Hot Standby Router Protocol)**
- **STP (Spanning Tree Protocol)**
- **NAT (Network Address Translation)**

Appendix E: Testing Screenshots

This appendix includes screenshots of:

- **Ping Test Results**
- **VLAN Communication Testing**
- **OSPF Routing Verification**
- **HSRP Failover Testing**
- **NAT Configuration and Internet Testing**

Appendix F: Configuration Commands (Optional)

Important Cisco configuration commands used in the project:

- VLAN Configuration Commands
- OSPF Configuration Commands
- HSRP Configuration Commands
- STP Configuration Commands
- NAT Configuration Commands

These appendices provide supporting details and additional information related to the project implementation and testing.